Catalog No. C14104

DADCO®

Ultra Force Extended® Nitrogen Gas Springs

UX Series



Patented

- **Reduce die construction costs**
- **Long stroke lengths – up to 300 mm**
- **Directly upgrade ISO 500-10000 models**
- **Variety of attachable and welded mounts**

PED
97/23/EC
COMPLIANT

Introduction

DADCO®
The global leader in nitrogen gas spring technology

Introduction

DADCO produces top quality products at competitive prices and provides a superior level of customer service. Founded in 1958, DADCO is the highest volume producer of gas springs for press tools. DADCO's products are widely approved and used in global operations for many industries including metal stamping, automotive, and plastic injection molding.



High Force + ISO Envelope = UX Series

DADCO's **Ultra Force Extended®** Series fits the standard full height gas spring envelope with more force on contact, making it ideal for use with high strength steel. It also means cost savings in die construction and maintenance. The UX Series is available in extended stroke lengths with a variety of welded and attachable mounts.

Model	Diameter	Maximum Force on Contact
New! UX.0800	44.5 mm *(1.750˝)*	7.3 kN *(0.7 ton)*
UX.1000 / UX.1000V	50 mm *(1.968˝)*	9.2 kN *(1.0 ton)*
UX.1600	63 mm *(2.480˝)*	15 kN *(1.7 ton)*
UX.2600 / UX.2600V	75 mm *(2.953˝)*	24 kN *(2.6 ton)*
UX.4600	95 mm *(3.740˝)*	42 kN *(4.7 ton)*
UX.6600	120 mm *(4.724˝)*	66 kN *(7.4 ton)*
UX.9600	150 mm *(5.906˝)*	95 kN *(10.7 ton)*
UX.20000	195 mm *(7.677˝)*	199 kN *(22 ton)*

Rapid Delivery

DADCO's modern 13,500 m^2 main production facility as well as satellite facilities permit the fastest deliveries in the industry. Products are available both directly and through a network of stocking distributors, providing worldwide support.

UltraPak® Technology

The UX Series uses the same superior **UltraPak®** cartridge as the popular **Ultra Force®** Series. The **UltraPak®** employs advanced materials to reduce the rod cartridge profile while increasing performance. The **UltraPak®** is comprised of a rod wiper, rod seal and guide arranged to extend life, retain lubrication, exclude contaminants, and provide excellent wear characteristics. When coupled with DADCO's superior finished piston rod, the **UltraPak®** is the ultimate sealing system.

Numerous Piping Options

With DADCO's **Ultra Force Extended®** Series, piping gas springs is easy because every necessary tool or component is readily available (e.g. reliable and cost-effective fittings, various hoses and couplings, control panels, solid fittings, crimping equipment). For additional information, refer to DADCO's Nitrogen Gas Spring Linked System Components Catalog.

SMS®

For those instances when a customer prefers to have DADCO provide a finished system, several options are available. A popular option, DADCO's Sectional Mounting System (SMS®) includes cylinders mounted to a SMS® plate and linked using hose, fittings and a control panel. Shipped ready to install, customers find the SMS® to be a trouble free, cost-effective option. For additional information, refer to catalog C13105.

SMS-i®

Another option is the Sectional Mounting System - Internal (SMS-i®). The SMS-i® consists of an internally piped plate with mounted cylinders. DADCO recommends using the SMS-i® as a cost effective alternative to traditional manifold systems. For more information, request catalog C13105.

Customer Satisfaction

DADCO's motto is *"Whatever It Takes To Satisfy Our Customers."* DADCO will assist in any way possible to ensure that customers are completely satisfied. DADCO's salespeople and distributors are solution-oriented, product-knowledgeable and eager to assist customers. DADCO's engineers are available to help customers with specific applications.

Warranty

DADCO warrants its UX Series Nitrogen Gas Springs to be free from defects in workmanship or materials for a period of one year from date of manufacture.

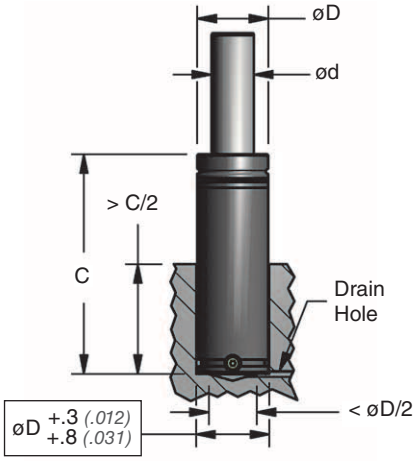
CAD Templates On-line

DADCO's entire product line is available on-line in solid models and 2D CAD formats. For more information, visit our website, www.dadco.net, or contact DADCO.

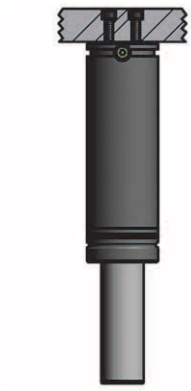
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Installation Examples

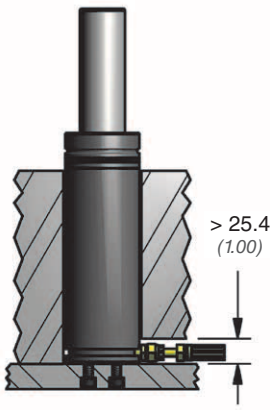
DADCO offers a variety of mount options to meet specific customer applications. Installation and fastening of the gas springs should take into consideration load support, fastener selection and torque values. For additional information on installation requirements see page 24. Cylinder and mount dimensions are shown on pages 4-21.



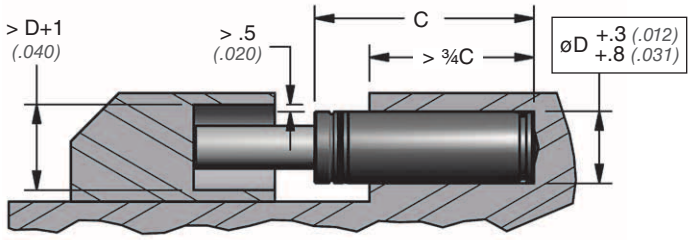
TO Basic Model in a flat bottom pocket. The pocket must be bored with a flat bottom, or a spacer must be used to create a flat surface.



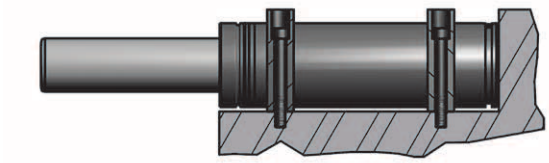
TO Basic Model mounted inverted requires back-up to support the full load. Retain inverted cylinders tight in the pocket with the appropriate length cap screw to eliminate movement.



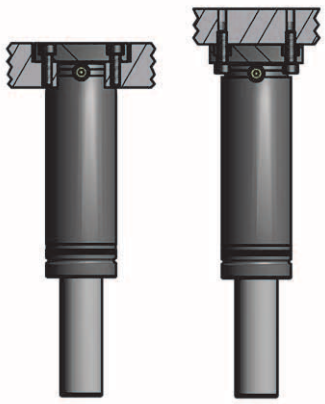
TO Basic Model mounted to a plate. Linked cylinders require clearance for the hose and fittings.



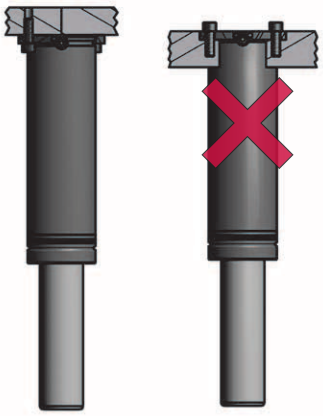
TO Basic Model mounted in a horizontal pocket. Allow clearance in the rod pocket to avoid contacting the body of the cylinder during operation. Allow the rod to locate freely.



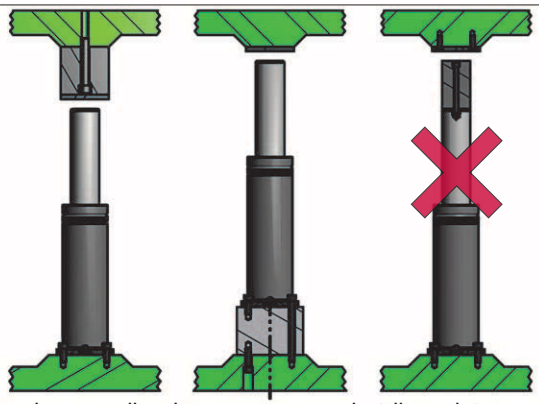
B19/B319V/B26 mounts require back-up to support the load. For long stroke models use two mounts to secure the cylinder.



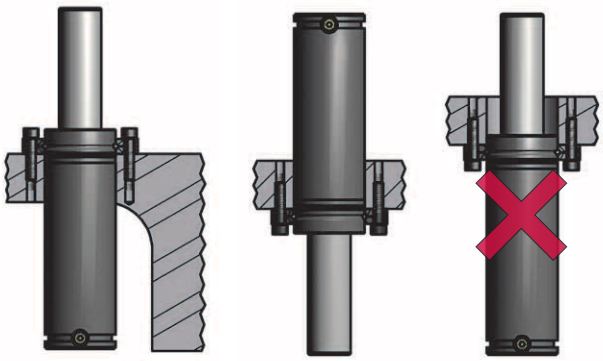
TFM/TSW/B11 mounts may be attached from the top or base. Back-up is required to support the full load. *NOTE: Mount 90.11.07500 from top only.*



B12 mounts must be fastened to the bottom groove only. Back-up is required to support the full load.

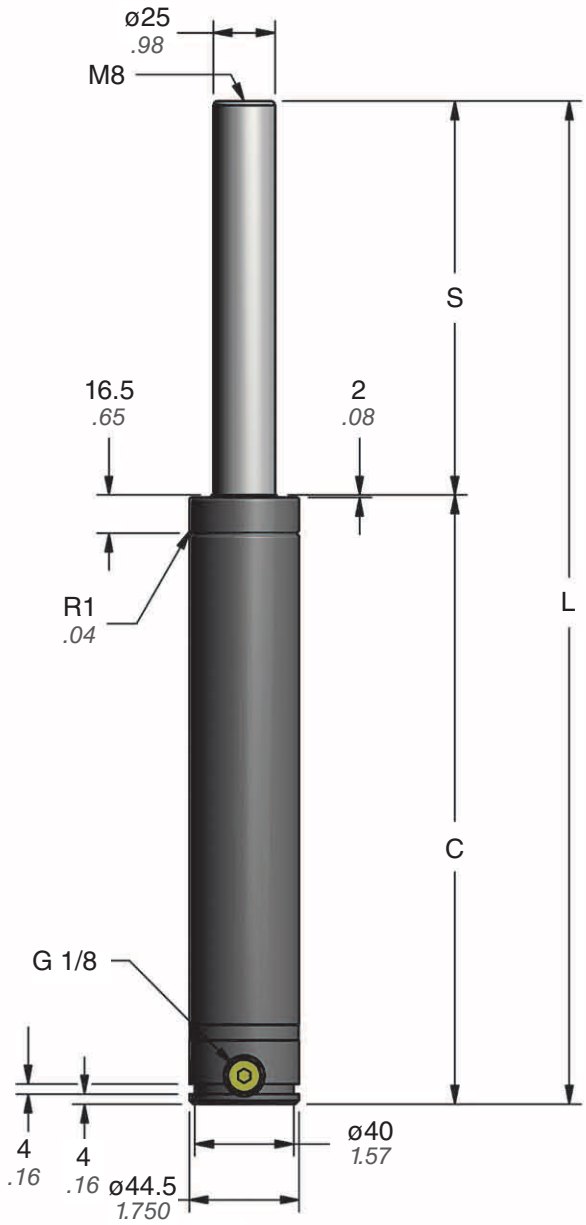


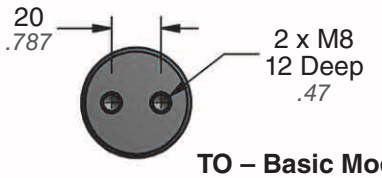
Pad extensions or die shoe spacers and striker plates may be used. Never secure anything to the rod end; the rod end has a construction thead intended only for assembly purposes.



B21/B25 mounts must be fastened to the top groove only. Back-up is required to support the full load. Do not mount from the bottom side of the mount.

UX.0800 – 7.3 kN / 0.7 ton *New!*





TO – Basic Model

Part No.	S mm *inch*	C	L ±0.25 *±.010*
UX.0800.013	12.5 *0.49*	97.5 *3.84*	110 *4.331*
• UX.0800.025	25 *0.98*	110 *4.33*	135 *5.315*
• UX.0800.038	37.5 *1.48*	122.5 *4.82*	160 *6.299*
• UX.0800.050	50 *1.97*	135 *5.31*	185 *7.283*
UX.0800.063	62.5 *2.46*	147.5 *5.81*	210 *8.268*
UX.0800.075	75 *2.95*	160 *6.30*	235 *9.252*
• UX.0800.080	80 *3.15*	165 *6.50*	245 *9.646*
• UX.0800.100	100 *3.94*	185 *7.28*	285 *11.220*
• UX.0800.125	125 *4.92*	210 *8.27*	335 *13.189*
UX.0800.150	150 *5.91*	235 *9.25*	385 *15.157*
• UX.0800.160	160 *6.30*	245 *9.65*	405 *15.945*
UX.0800.175	175 *6.89*	260 *10.24*	435 *17.126*
UX.0800.200	200 *7.87*	285 *11.22*	485 *19.094*

• Preferred Sizes

Ordering Example:

UX.0800.080. *TO.* *C.* *150*

Part Number:
Includes Series, Model and Stroke Length.

Mount Option:
TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

Mount Only Ordering Example: 90.21.00500

Charging Pressure:
15–150 bar *(220–2175 psi). When not specified, default is 150 bar.*

Operating System:
C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

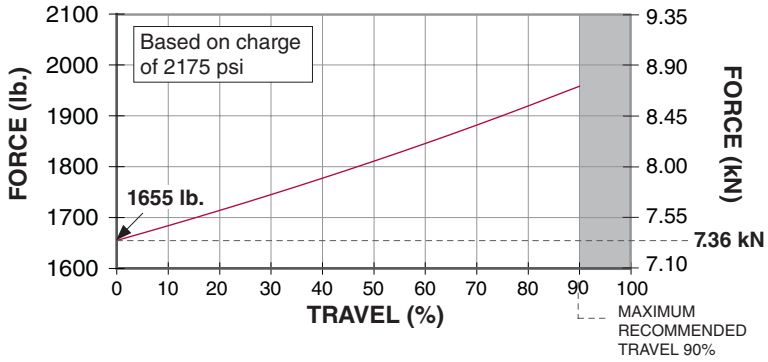
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UX.0800 – 7.3 kN / 0.7 ton

Force Charts

On-Contact Force

psi	lb.
2175	1655
2000	1522
1750	1331
1500	1141
1000	761
500	380
250	190

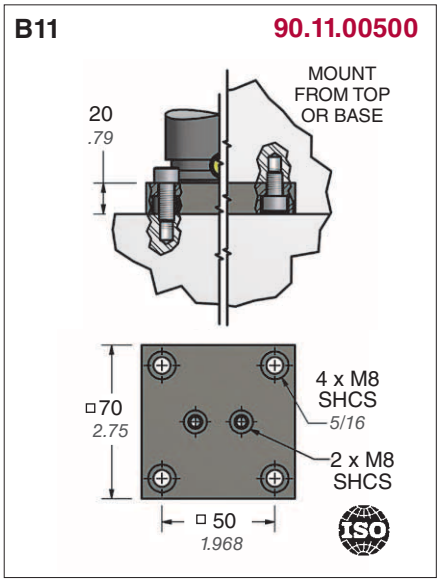


On-Contact Force

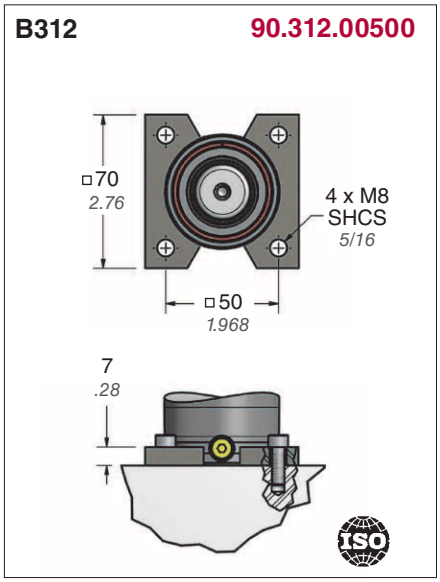
bar	kN
150	7.36
125	6.14
100	4.91
75	3.68
50	2.45
25	1.23
20	0.98

Mount Options

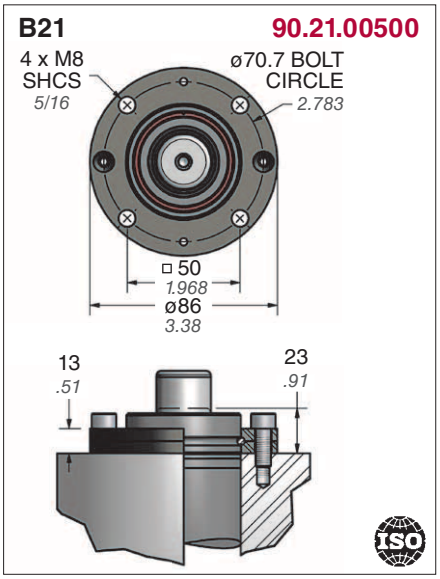
B11 **90.11.00500**



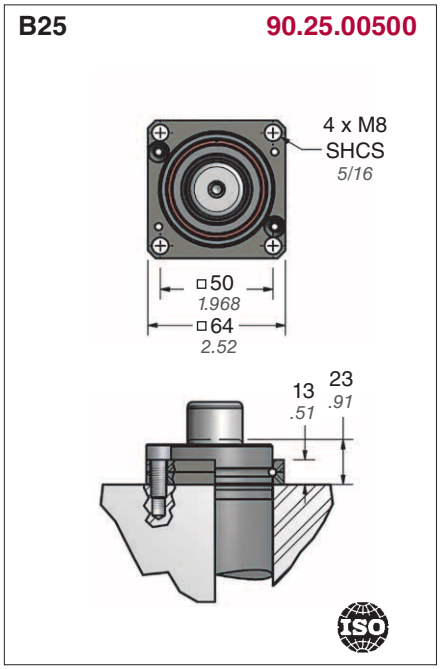
B312 **90.312.00500**



B21 **90.21.00500**

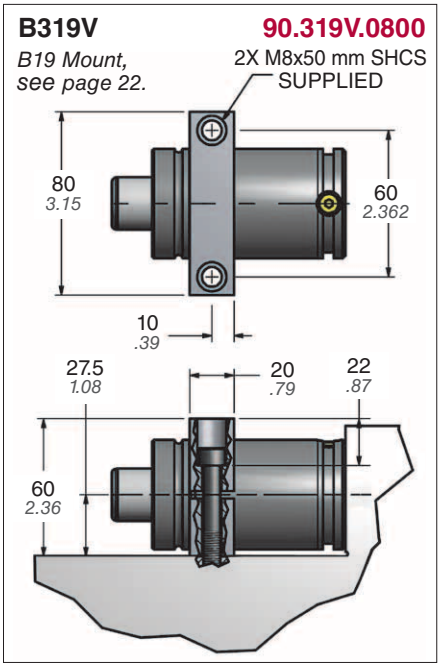


B25 **90.25.00500**



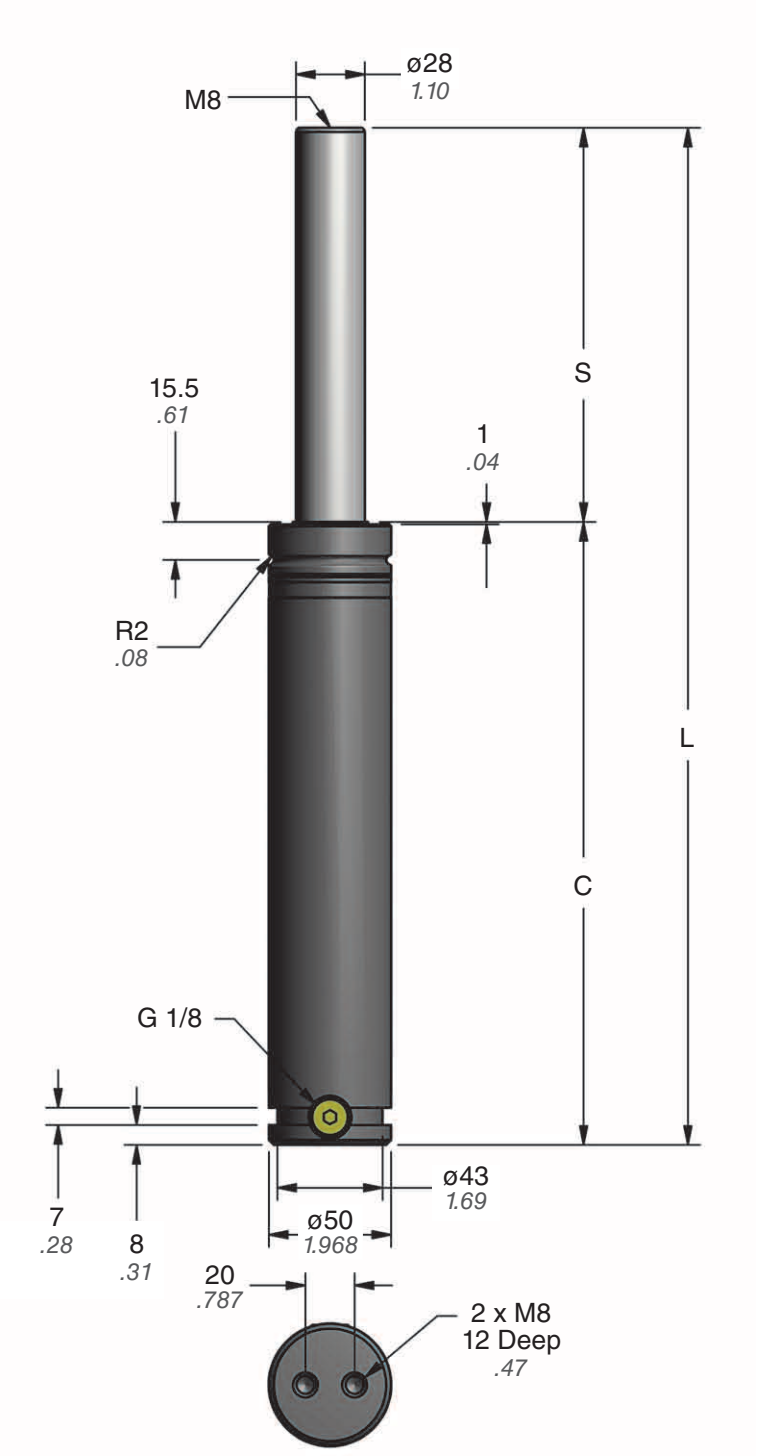
B319V **90.319V.0800**

B19 Mount, see page 22.



DADCO offers a wide variety of mounts. If the mounts above are not suitable for the application, contact DADCO.

UX.1000 – 9.2 kN / 1.0 ton



TO – Basic Model

Refer to page 20 for model matching European Standards.

Part No.	S mm *inch*	C	L ±0.25 *±.010*
UX.1000.013	12.5 *.49*	105.5 *4.15*	118 *4.646*
• UX.1000.025	25 *0.98*	118 *4.65*	143 *5.630*
• UX.1000.038	37.5 *1.48*	130.5 *5.14*	168 *6.614*
• UX.1000.050	50 *1.97*	143 *5.63*	193 *7.598*
UX.1000.063	62.5 *2.46*	155.5 *6.12*	218 *8.583*
UX.1000.075	75 *2.95*	168 *6.61*	243 *9.567*
• UX.1000.080	80 *3.15*	173 *6.81*	253 *9.961*
• UX.1000.100	100 *3.94*	193 *7.60*	293 *11.535*
• UX.1000.125	125 *4.92*	218 *8.58*	343 *13.504*
UX.1000.150	150 *5.91*	243 *9.57*	393 *15.472*
• UX.1000.160	160 *6.30*	253 *9.96*	413 *16.260*
UX.1000.175	175 *6.89*	268 *10.55*	443 *17.441*
UX.1000.200	200 *7.87*	293 *11.54*	493 *19.409*
UX.1000.225	225 *8.86*	318 *12.52*	543 *21.378*
UX.1000.250	250 *9.84*	343 *13.50*	593 *23.346*
UX.1000.275	275 *10.83*	368 *14.49*	643 *25.315*
UX.1000.300	300 *11.81*	393 *15.47*	693 *27.283*

• Preferred Sizes

Ordering Example:

UX.1000.080. *TO.* *C.* *150*

Part Number:
Includes Series, Model and Stroke Length.

Mount Option:
TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

Mount Only Ordering Example: 90.21.00750

Charging Pressure:
15–150 bar *(220–2175 psi). When not specified, default is 150 bar.*

Operating System:
C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

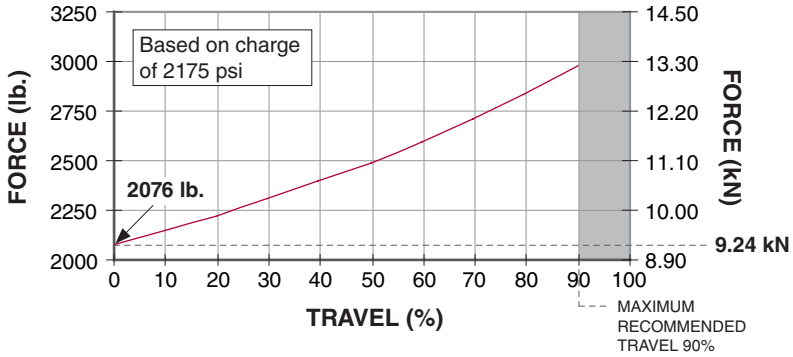
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UX.1000 – 9.2 kN / 1.0 ton

Force Charts

On-Contact Force

psi	lb.
2175	2076
2000	1909
1750	1670
1500	1432
1000	954
500	477
250	239

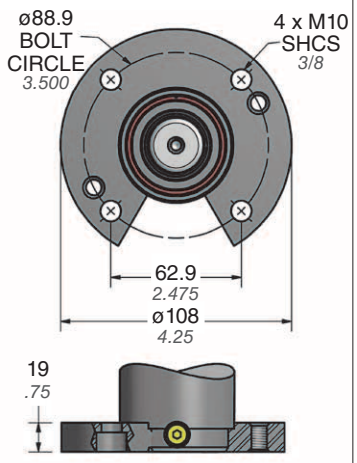


On-Contact Force

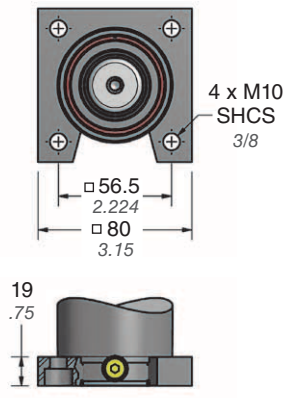
bar	kN
150	9.24
125	7.70
100	6.16
75	4.62
50	3.08
25	1.54
20	1.23

Mount Options

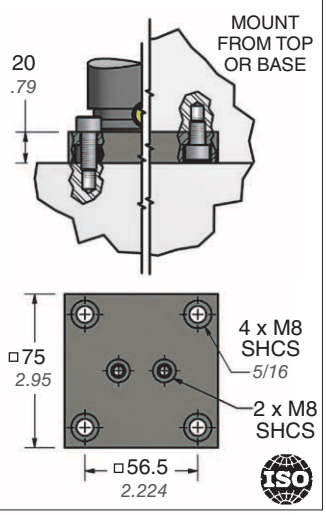
TFM Welded



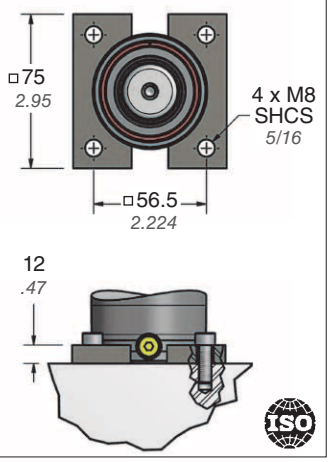
TSW Welded



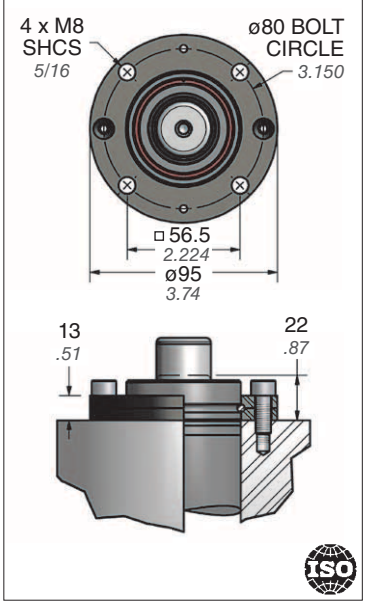
B11 90.11.00750



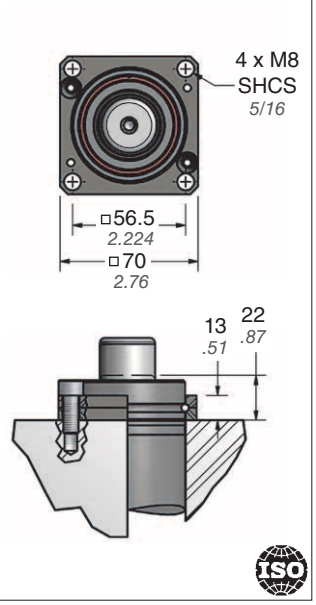
B12 90.12.00750



B21 90.21.00750

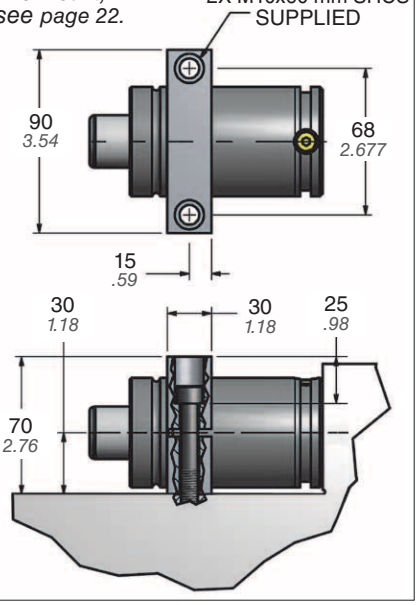


B25 90.25.00750



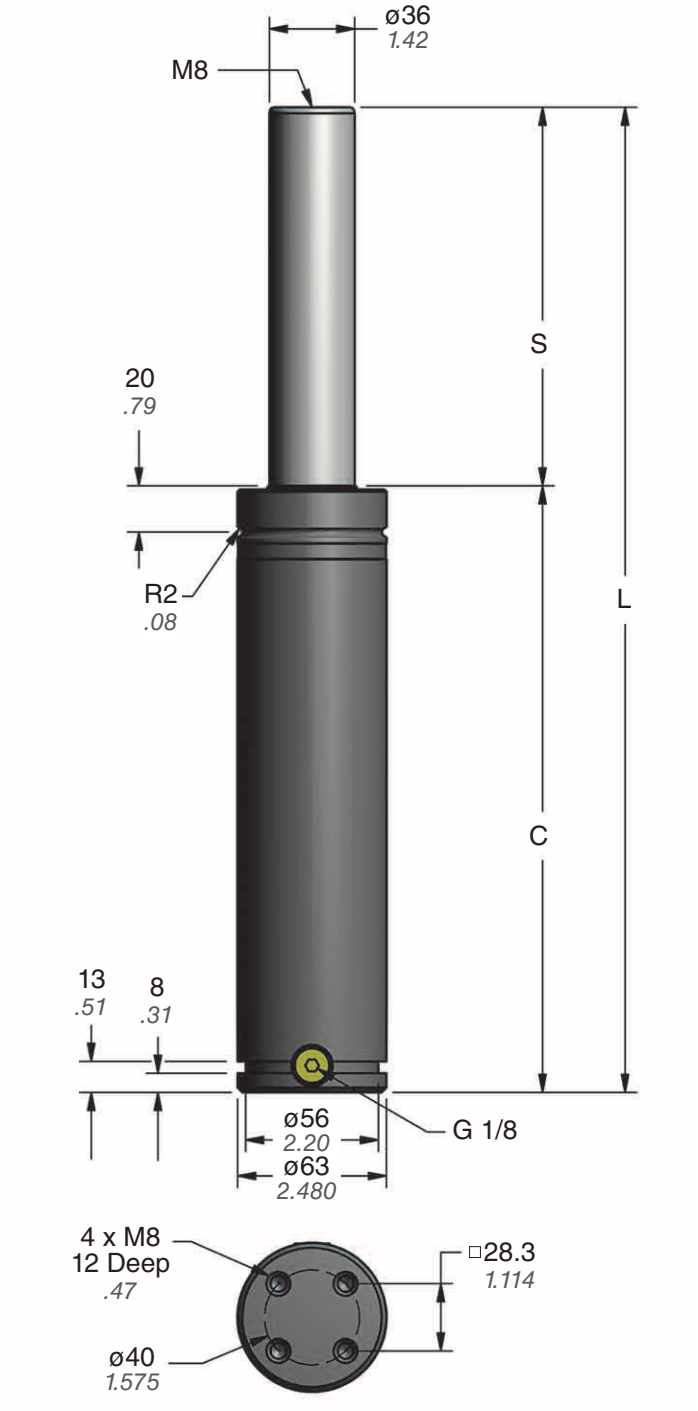
B319V 90.319V.1000

B19 Mount, see page 22.



DADCO offers a wide variety of mounts. If the mounts above are not suitable for the application, contact DADCO.

UX.1600 – 15 kN / 1.7 ton



TO – Basic Model

Part No.	S mm *inch*	C	L ±0.25 *±.010*
UX.1600.013	12.5 *0.49*	107.5 *4.23*	120 *4.724*
• UX.1600.025	25 *0.98*	120 *4.72*	145 *5.709*
• UX.1600.038	37.5 *1.48*	132.5 *5.22*	170 *6.693*
• UX.1600.050	50 *1.97*	145 *5.71*	195 *7.677*
UX.1600.063	62.5 *2.46*	157.5 *6.20*	220 *8.661*
UX.1600.075	75 *2.95*	170 *6.69*	245 *9.646*
• UX.1600.080	80 *3.15*	175 *6.89*	255 *10.039*
• UX.1600.100	100 *3.94*	195 *7.68*	295 *11.614*
• UX.1600.125	125 *4.92*	220 *8.66*	345 *13.583*
UX.1600.150	150 *5.91*	245 *9.65*	395 *15.551*
• UX.1600.160	160 *6.30*	255 *10.04*	415 *16.339*
UX.1600.175	175 *6.89*	270 *10.63*	445 *17.520*
UX.1600.200	200 *7.87*	295 *11.61*	495 *19.488*
UX.1600.225	225 *8.86*	320 *12.60*	545 *21.457*
UX.1600.250	250 *9.84*	345 *13.58*	595 *23.425*
UX.1600.275	275 *10.83*	370 *14.57*	645 *25.394*
UX.1600.300	300 *11.81*	395 *15.55*	695 *27.362*

• Preferred Sizes

Ordering Example:

UX.1600.080. *TO*. *C*. *150*

Part Number:
Includes Series, Model and Stroke Length.

Mount Option:
TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

Mount Only Ordering Example: 90.21.1600

Charging Pressure:
15–150 bar *(220–2175 psi). When not specified, default is 150 bar.*

Operating System:
C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

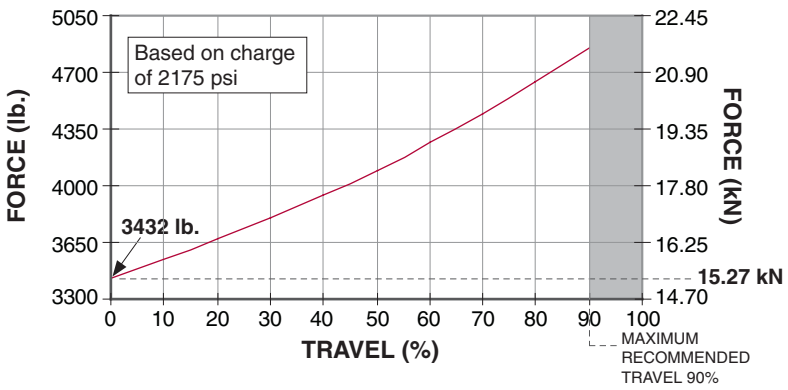
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UX.1600 – 15 kN / 1.7 ton

Force Charts

On-Contact Force

psi	lb.
2175	3432
2000	3155
1750	2761
1500	2367
1000	1578
500	789
250	394

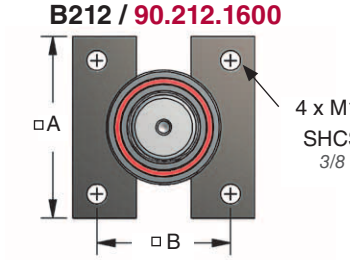


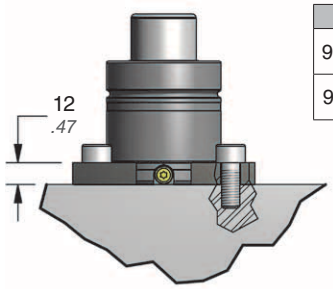
On-Contact Force

bar	kN
150	15.27
125	12.72
100	10.18
75	7.63
50	5.09
25	2.54
20	2.04

Mount Options

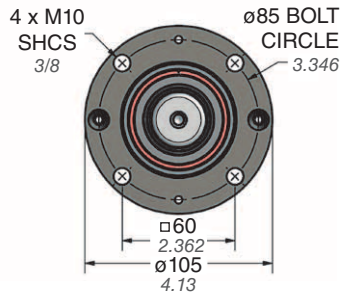
B212 / 90.212.1600 — B112 / 90.112.1600

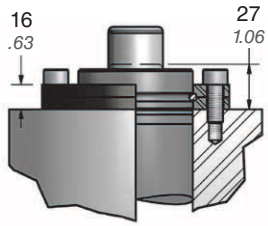




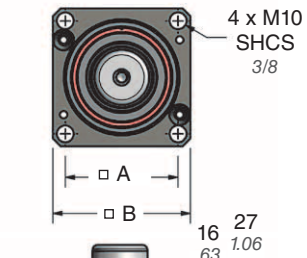
Part No.	A	B
90.212.1600	100 *3.94*	73.5 *2.894*
90.112.1600	85 *3.35*	60 *2.362*

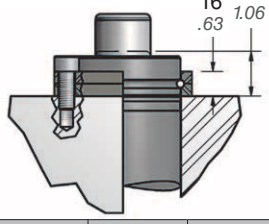
B21 — 90.21.1600





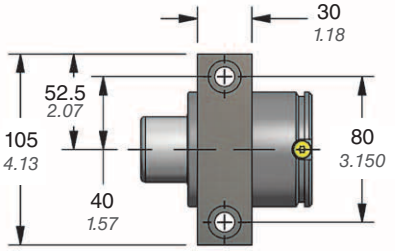
B325/B25 — 90.325.1600/ 90.25.1600

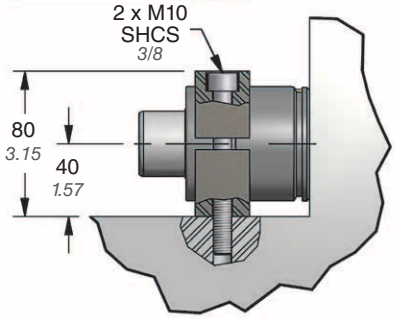




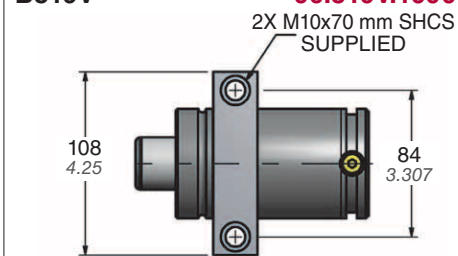
Part No.	A	B
90.325.1600	64 *2.520*	80 *3.54*
90.25.1600	73.5 *2.894*	90 *3.54*

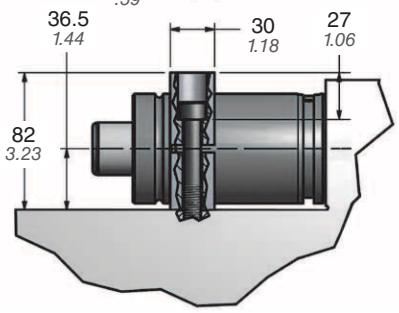
B319 — 90.319.1600





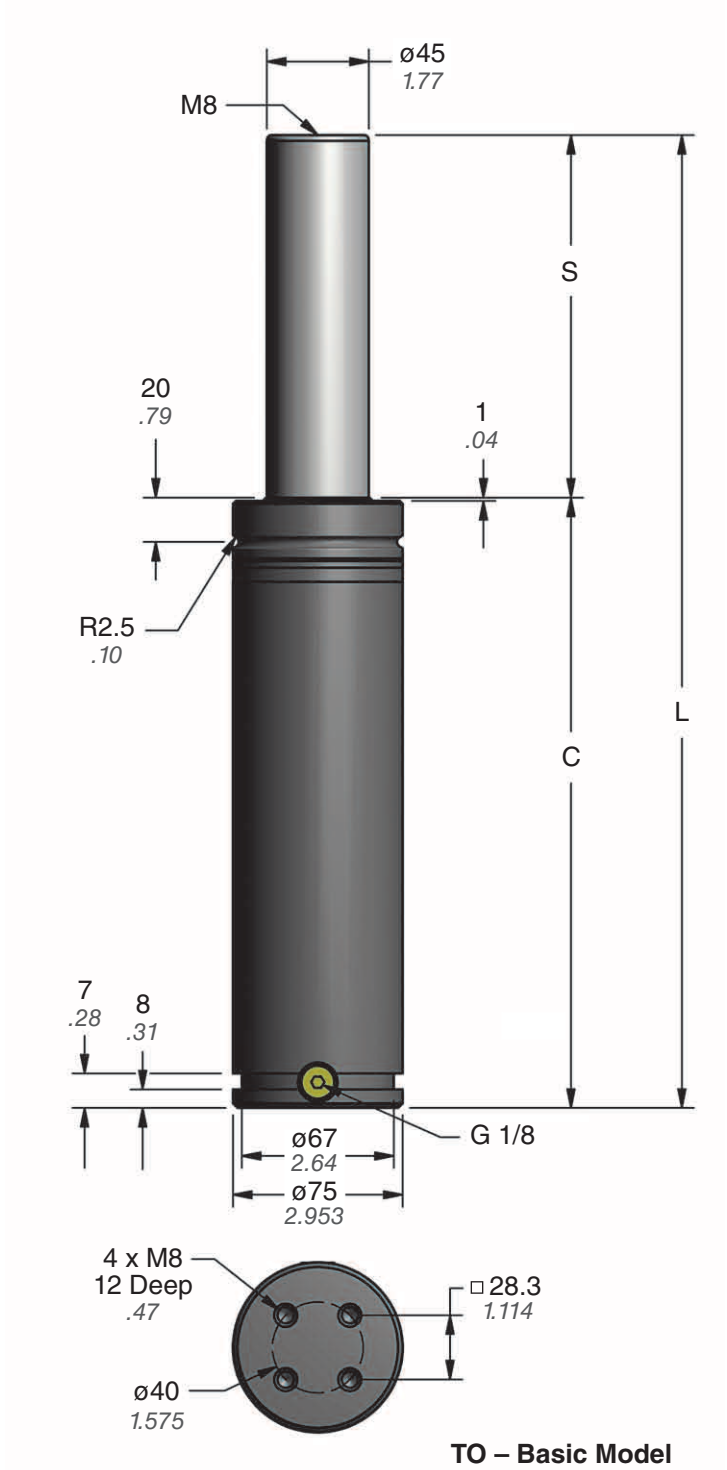
B319V — 90.319V.1600





DADCO offers a wide variety of mounts. If the mounts above are not suitable for the application, contact DADCO.

UX.2600 – 24 kN / 2.6 ton



TO – Basic Model

Refer to page 21 for model matching European Standards.

Part No.	S mm *inch*	C	L ±0.25 *±.010*
• UX.2600.025	25 *0.98*	133 *5.24*	158 *6.220*
• UX.2600.038	37.5 *1.48*	145.5 *5.73*	183 *7.205*
• UX.2600.050	50 *1.97*	158 *6.22*	208 *8.189*
UX.2600.063	62.5 *2.46*	170.5 *6.71*	233 *9.173*
UX.2600.075	75 *2.95*	183 *7.20*	258 *10.157*
• UX.2600.080	80 *3.15*	188 *7.40*	268 *10.551*
• UX.2600.100	100 *3.94*	208 *8.19*	308 *12.126*
• UX.2600.125	125 *4.92*	233 *9.17*	358 *14.094*
UX.2600.150	150 *5.91*	258 *10.16*	408 *16.063*
• UX.2600.160	160 *6.30*	268 *10.55*	428 *16.850*
UX.2600.175	175 *6.89*	283 *11.14*	458 *18.031*
UX.2600.200	200 *7.87*	308 *12.13*	508 *20.000*
UX.2600.225	225 *8.86*	333 *13.11*	558 *21.968*
UX.2600.250	250 *9.84*	358 *14.09*	608 *23.937*
UX.2600.275	275 *10.83*	383 *15.08*	658 *25.906*
UX.2600.300	300 *11.81*	408 *16.06*	708 *27.874*

• Preferred Sizes

Ordering Example:

UX.2600.080. *TO*. *C*. *150*

Part Number:
Includes Series, Model and Stroke Length.

Mount Option:
TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

Mount Only Ordering Example: 90.21.01500

Charging Pressure:
15–150 bar *(220–2175 psi). When not specified, default is 150 bar.*

Operating System:
C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

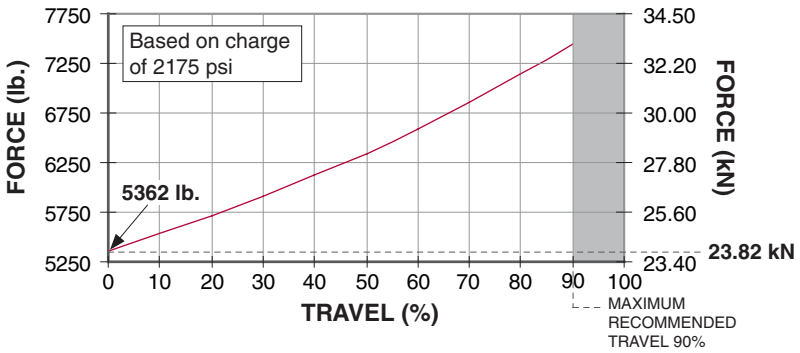
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UX.2600 – 24 kN / 2.6 ton

Force Charts

On-Contact Force

psi	lb.
2175	5362
2000	4930
1750	4314
1500	3698
1000	2465
500	1233
250	616

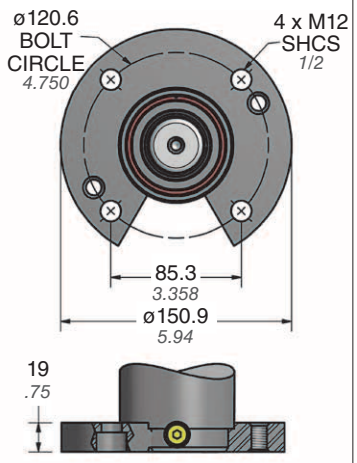


On-Contact Force

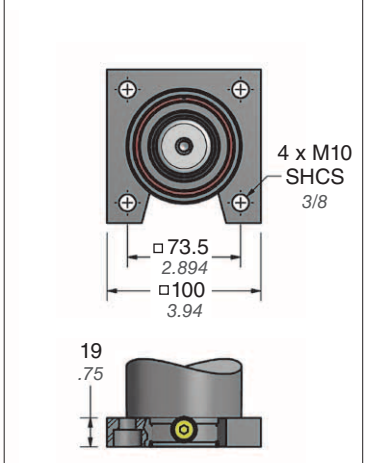
bar	kN
150	23.86
125	19.88
100	15.90
75	11.93
50	7.95
25	3.98
20	3.18

Mount Options

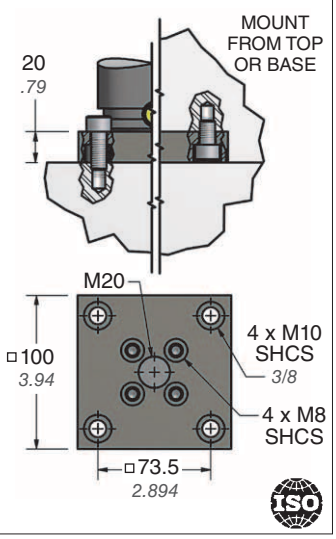
TFM Welded



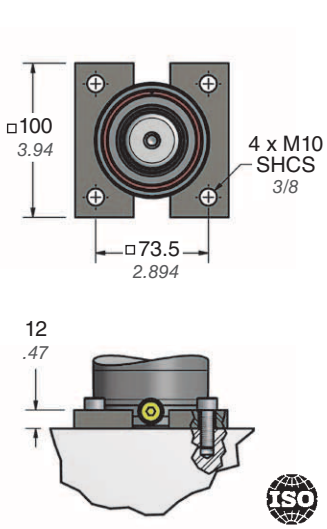
TSW Welded



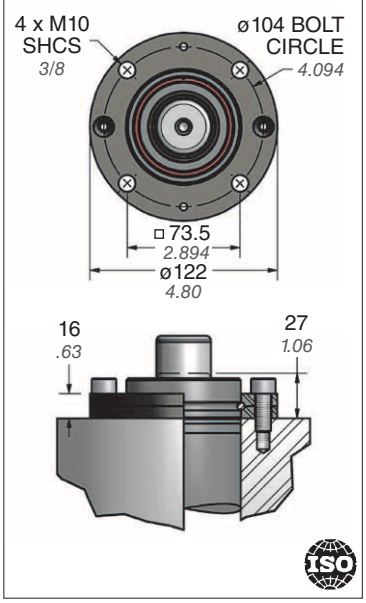
B11 90.11.01500



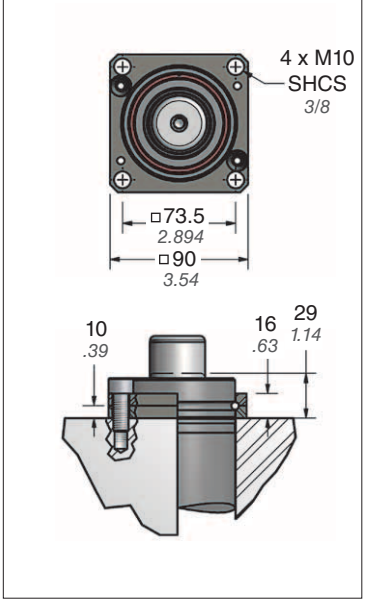
B12 90.12.01500



B21 90.21.01500

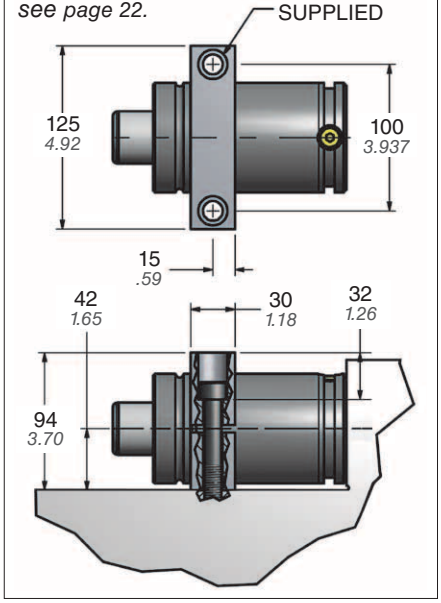


B325 90.325.2600



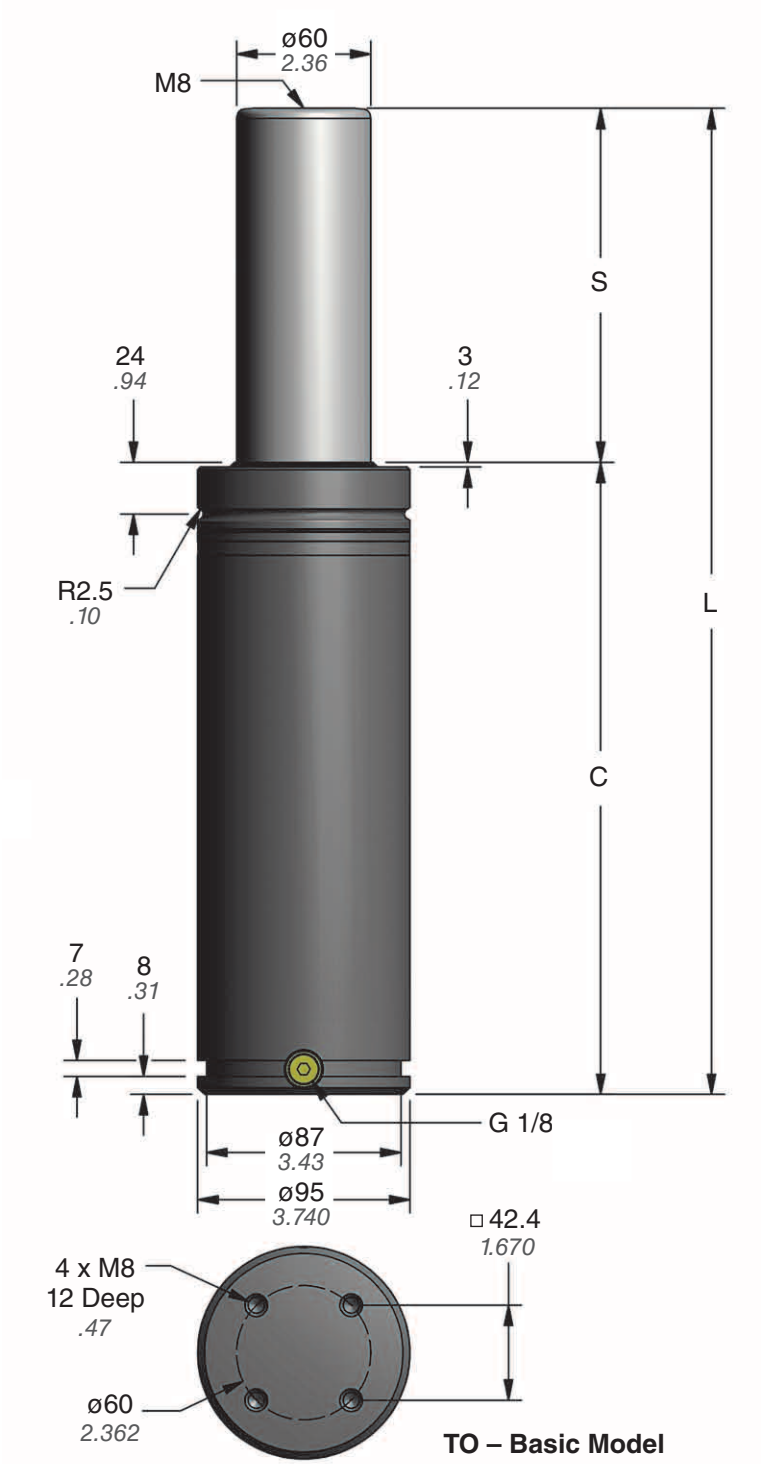
B319V 90.319V.2600

B19 Mount, see page 22.



DADCO offers a wide variety of mounts. If the mounts above are not suitable for the application, contact DADCO.

UX.4600 – 42 kN / 4.7 ton



TO – Basic Model

Part No.	S mm *inch*	C	L ±0.25 *±.010*
• UX.4600.025	25 *0.98*	145 *5.71*	170 *6.693*
• UX.4600.038	37.5 *1.48*	157.5 *6.20*	195 *7.677*
• UX.4600.050	50 *1.97*	170 *6.69*	220 *8.661*
UX.4600.063	62.5 *2.46*	182.5 *7.19*	245 *9.646*
UX.4600.075	75 *2.95*	195 *7.68*	270 *10.630*
• UX.4600.080	80 *3.15*	200 *7.87*	280 *11.024*
• UX.4600.100	100 *3.94*	220 *8.66*	320 *12.598*
• UX.4600.125	125 *4.92*	245 *9.65*	370 *14.567*
UX.4600.150	150 *5.91*	270 *10.63*	420 *16.535*
• UX.4600.160	160 *6.30*	280 *11.02*	440 *17.323*
UX.4600.175	175 *6.89*	295 *11.61*	470 *18.504*
UX.4600.200	200 *7.87*	320 *12.60*	520 *20.472*
UX.4600.225	225 *8.86*	345 *13.58*	570 *22.441*
UX.4600.250	250 *9.84*	370 *14.57*	620 *24.409*
UX.4600.275	275 *10.83*	395 *15.55*	670 *26.378*
UX.4600.300	300 *11.81*	420 *16.54*	720 *28.346*

• Preferred Sizes

Ordering Example:

UX.4600.080. *TO.* *C.* *150*

Part Number:
Includes Series, Model and Stroke Length.

Mount Option:
TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

Mount Only Ordering Example: 90.21.03000

Charging Pressure:
15–150 bar *(220–2175 psi). When not specified, default is 150 bar.*

Operating System:
C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

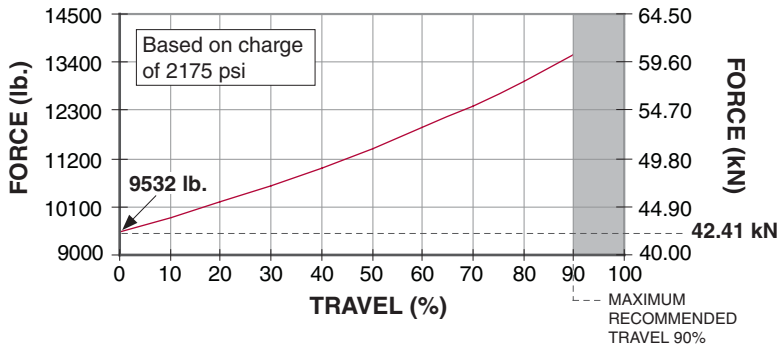
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UX.4600 – 42 kN / 4.7 ton

Force Charts

On-Contact Force

psi	lb.
2175	9532
2000	8765
1750	7669
1500	6574
1000	4383
500	2191
250	1096

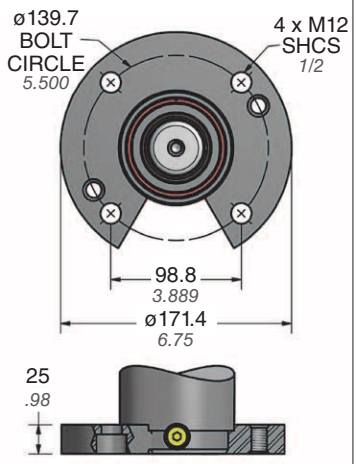


On-Contact Force

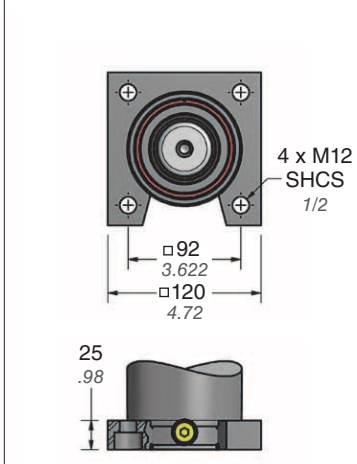
bar	kN
150	42.41
125	35.34
100	28.27
75	21.21
50	14.14
25	7.07
20	5.65

Mount Options

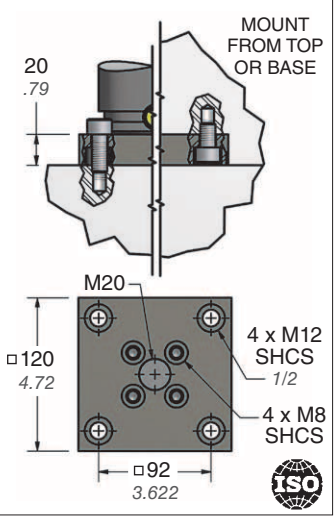
TFM — Welded



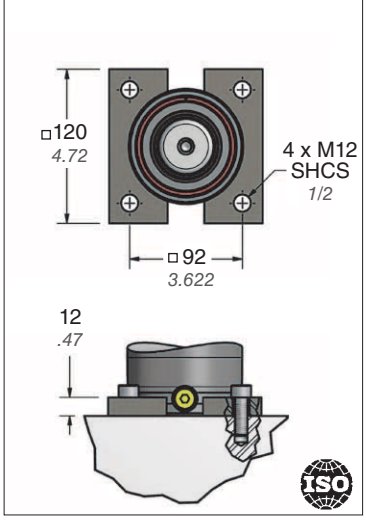
TSW — Welded



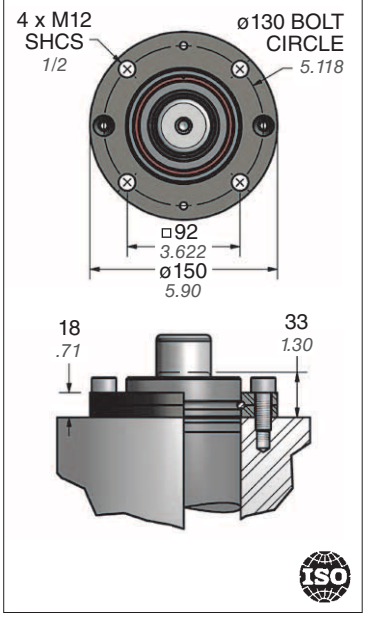
B11 — 90.11.03000



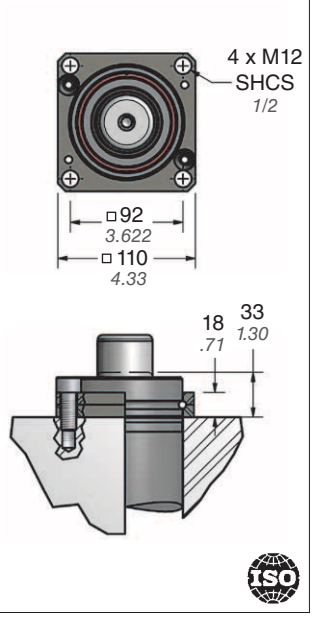
B12 — 90.12.03000



B21 — 90.21.03000

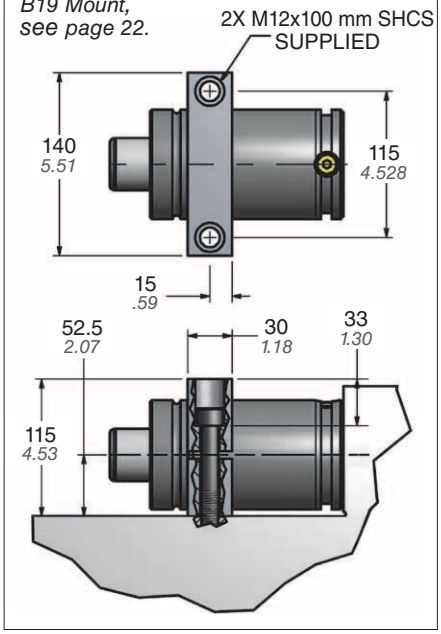


B25 — 90.25.03000



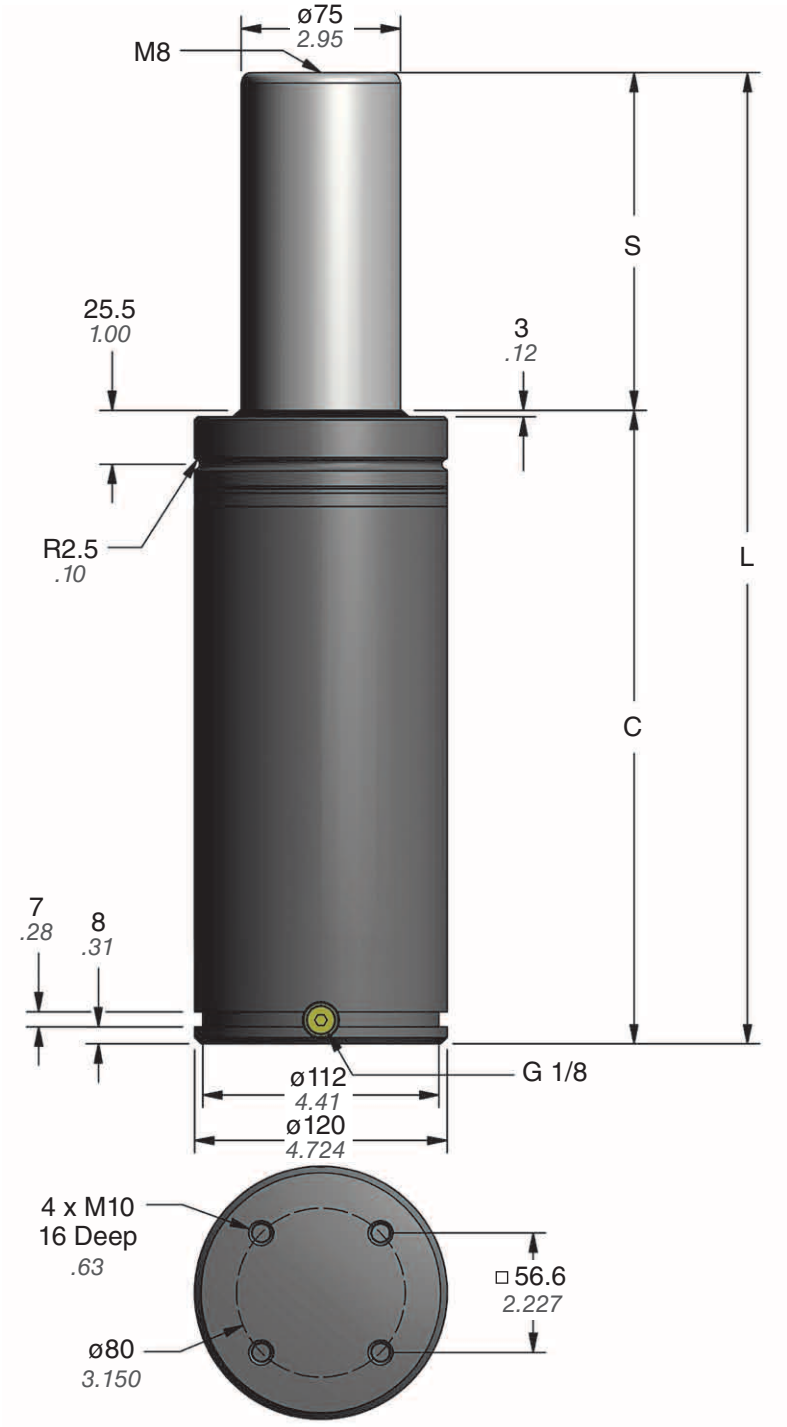
B319V — 90.319V.4600

B19 Mount, see page 22.



DADCO offers a wide variety of mounts. If the mounts above are not suitable for the application, contact DADCO.

UX.6600 – 66 kN / 7.4 ton



TO – Basic Model

Part No.	S mm *inch*	C	L ±0.25 *±.010*
• UX.6600.025	25 *0.98*	165 *6.50*	190 *7.480*
• UX.6600.038	37.5 *1.48*	177.5 *6.99*	215 *8.465*
• UX.6600.050	50 *1.97*	190 *7.48*	240 *9.449*
UX.6600.063	62.5 *2.46*	202.5 *7.97*	265 *10.433*
UX.6600.075	75 *2.95*	215 *8.46*	290 *11.417*
• UX.6600.080	80 *3.15*	220 *8.66*	300 *11.811*
• UX.6600.100	100 *3.94*	240 *9.45*	340 *13.386*
• UX.6600.125	125 *4.92*	265 *10.43*	390 *15.354*
UX.6600.150	150 *5.91*	290 *11.42*	440 *17.323*
• UX.6600.160	160 *6.30*	300 *11.81*	460 *18.110*
UX.6600.175	175 *6.89*	315 *12.40*	490 *19.291*
UX.6600.200	200 *7.87*	340 *13.39*	540 *21.260*
UX.6600.225	225 *8.86*	365 *14.37*	590 *23.228*
UX.6600.250	250 *9.84*	390 *15.35*	640 *25.197*
UX.6600.275	275 *10.83*	415 *16.34*	690 *27.165*
UX.6600.300	300 *11.81*	440 *17.32*	740 *29.134*

• Preferred Sizes

Ordering Example:

UX.6600.080. *TO.* *C.* *150*

Part Number:
Includes Series, Model and Stroke Length.

Mount Option:
TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

Mount Only Ordering Example: 90.21.05000

Charging Pressure:
15–150 bar *(220–2175 psi). When not specified, default is 150 bar.*

Operating System:
C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

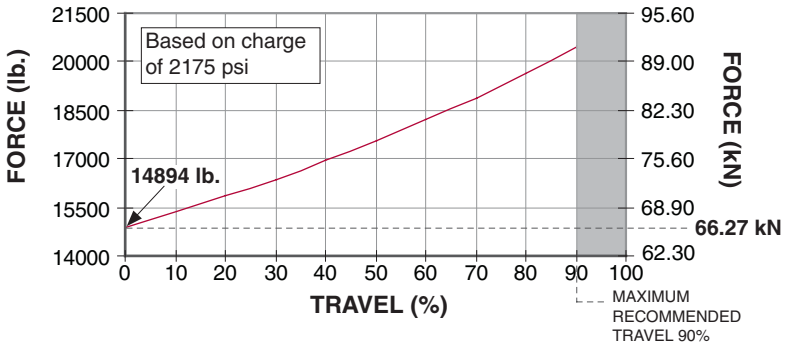
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UX.6600 – 66 kN / 7.4 ton

Force Charts

On-Contact Force

psi	lb.
2175	14894
2000	13695
1750	11983
1500	10272
1000	6848
500	3424
250	1712

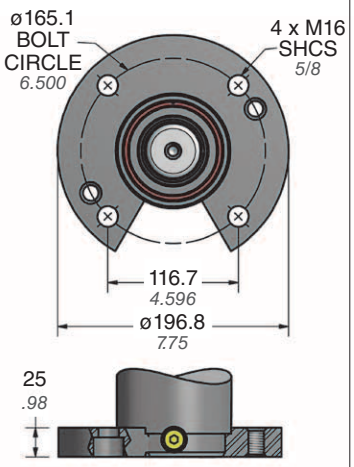


On-Contact Force

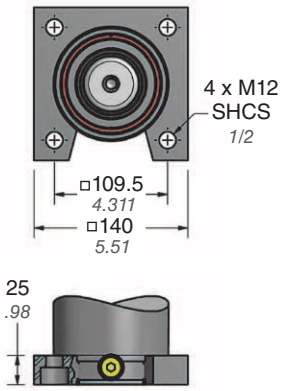
bar	kN
150	66.27
125	55.22
100	44.18
75	33.13
50	22.09
25	11.04
20	8.84

Mount Options

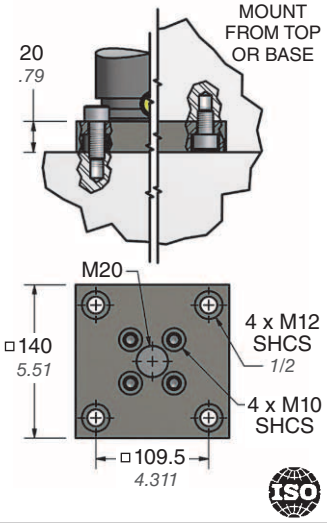
TFM Welded



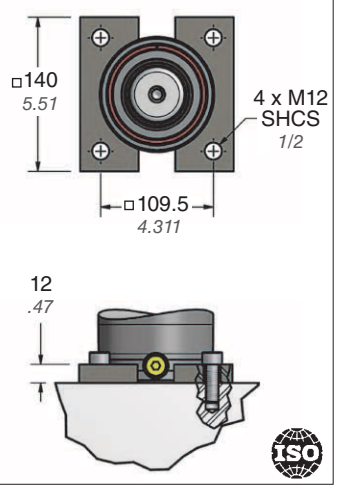
TSW Welded



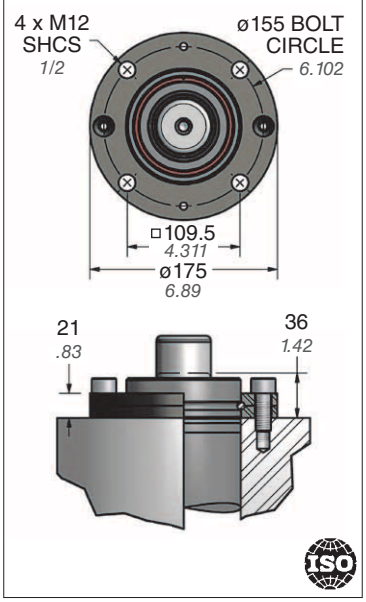
B11 90.11.05000



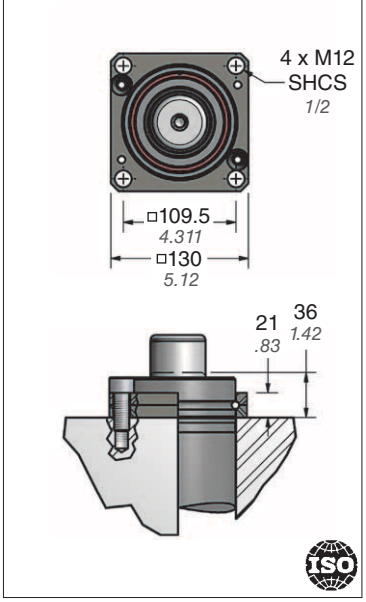
B12 90.12.05000



B21 90.21.05000

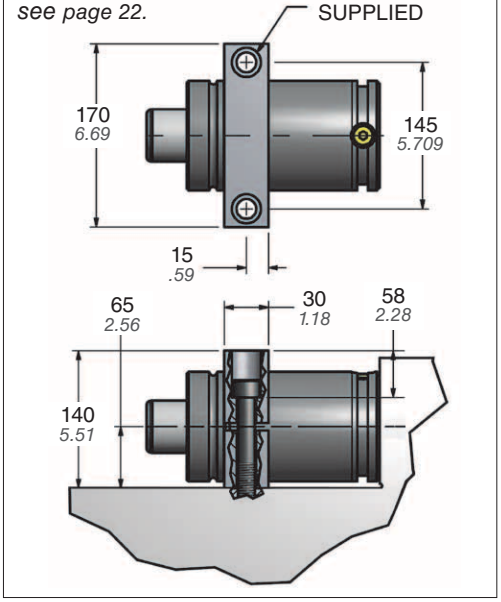


B25 90.25.05000



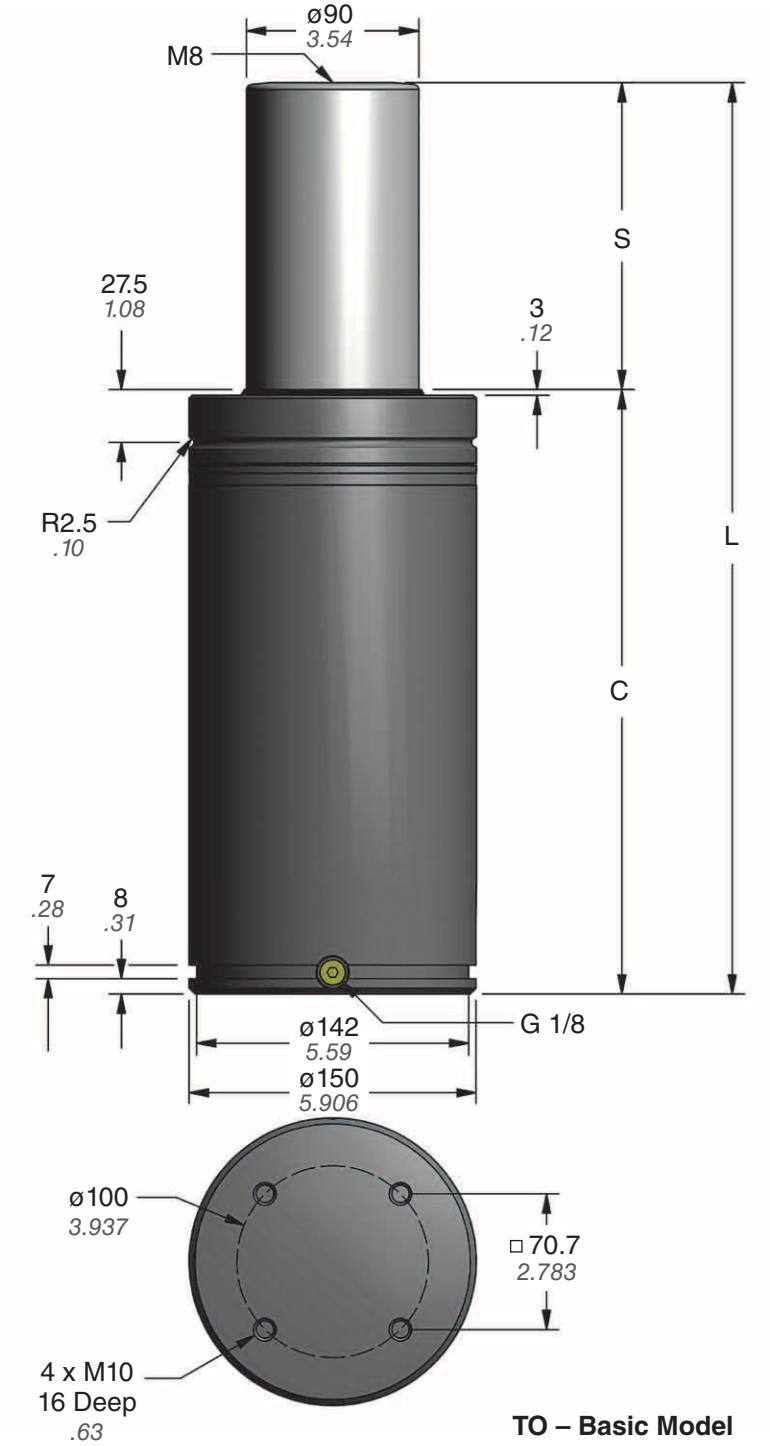
B319V 90.319V.6600

B19 Mount, see page 22.



DADCO offers a wide variety of mounts. If the mounts above are not suitable for the application, contact DADCO.

UX.9600 – 95 kN / 10.7 ton



TO – Basic Model

Part No.	S mm *inch*	C	L ±0.25 *±.010*
• UX.9600.025	25 *0.98*	180 *7.09*	205 *8.071*
UX.9600.038	37.5 *1.48*	192.5 *7.58*	230 *9.055*
• UX.9600.050	50 *1.97*	205 *8.07*	255 *10.039*
UX.9600.063	62.5 *2.46*	217.5 *8.56*	280 *11.024*
UX.9600.075	75 *2.95*	230 *9.06*	305 *12.008*
• UX.9600.080	80 *3.15*	235 *9.25*	315 *12.402*
• UX.9600.100	100 *3.94*	255 *10.04*	355 *13.976*
• UX.9600.125	125 *4.92*	280 *11.02*	405 *15.945*
UX.9600.150	150 *5.91*	305 *12.01*	455 *17.913*
• UX.9600.160	160 *6.30*	315 *12.40*	475 *18.701*
UX.9600.175	175 *6.89*	330 *12.99*	505 *19.882*
UX.9600.200	200 *7.87*	355 *13.98*	555 *21.850*
UX.9600.225	225 *8.86*	380 *14.96*	605 *23.819*
UX.9600.250	250 *9.84*	405 *15.94*	655 *25.787*
UX.9600.275	275 *10.83*	430 *16.93*	705 *27.756*
UX.9600.300	300 *11.81*	455 *17.91*	755 *29.724*

• Preferred Sizes

Ordering Example:

UX.9600.080. *TO.* *C.* *150*

Part Number:
Includes Series, Model and Stroke Length.

Mount Option:
TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

Mount Only Ordering Example: 90.21.07500

Charging Pressure:
15–150 bar *(220–2175 psi). When not specified, default is 150 bar.*

Operating System:
C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

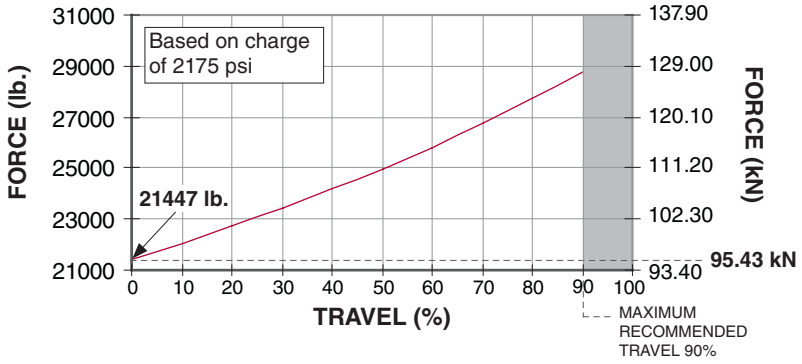
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UX.9600 – 95 kN / 10.7 ton

Force Charts

On-Contact Force

psi	lb.
2175	21447
2000	19721
1750	17256
1500	14791
1000	9861
500	4930
250	2465

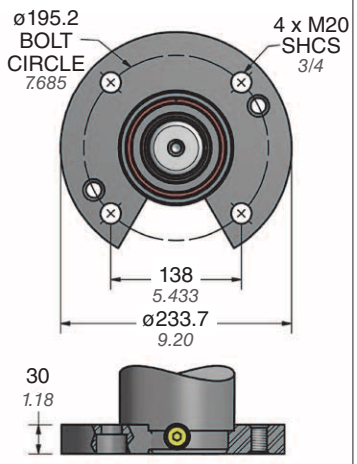


On-Contact Force

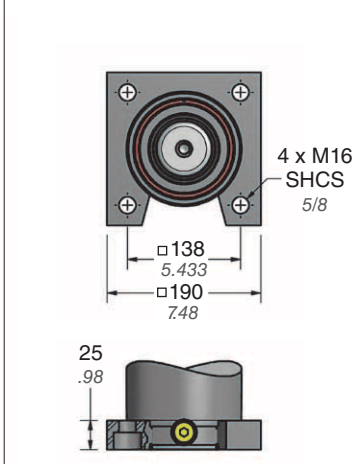
bar	kN
150	95.43
125	79.52
100	63.62
75	47.71
50	31.81
25	15.90
20	12.72

Mount Options

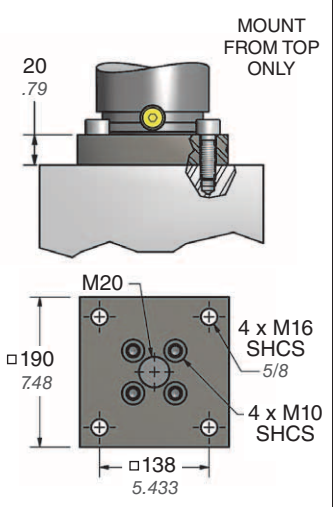
TFM Welded



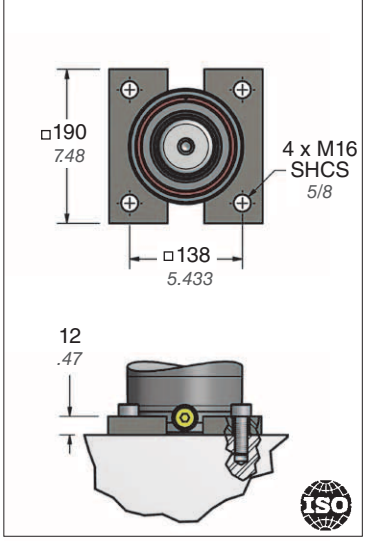
TSW Welded



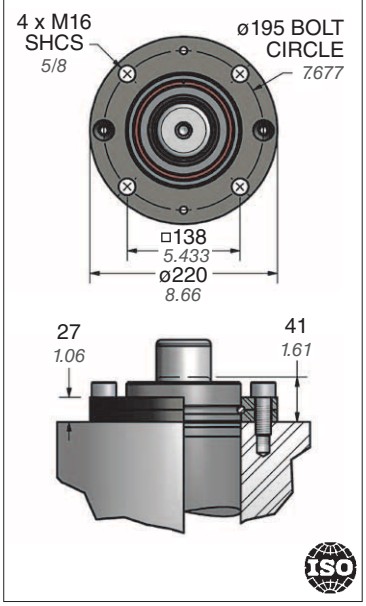
B11 90.11.07500



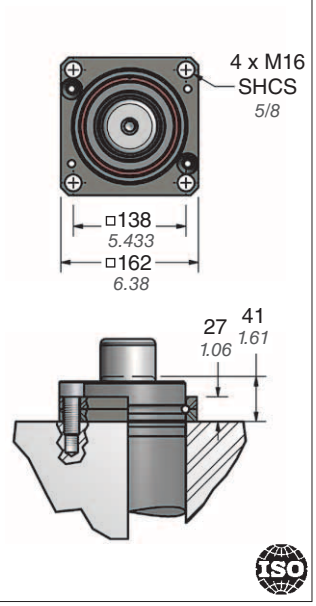
B12 90.12.07500



B21 90.21.07500

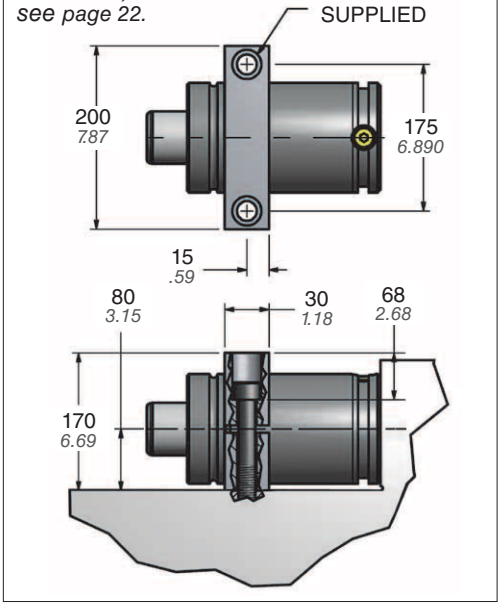


B25 90.25.07500



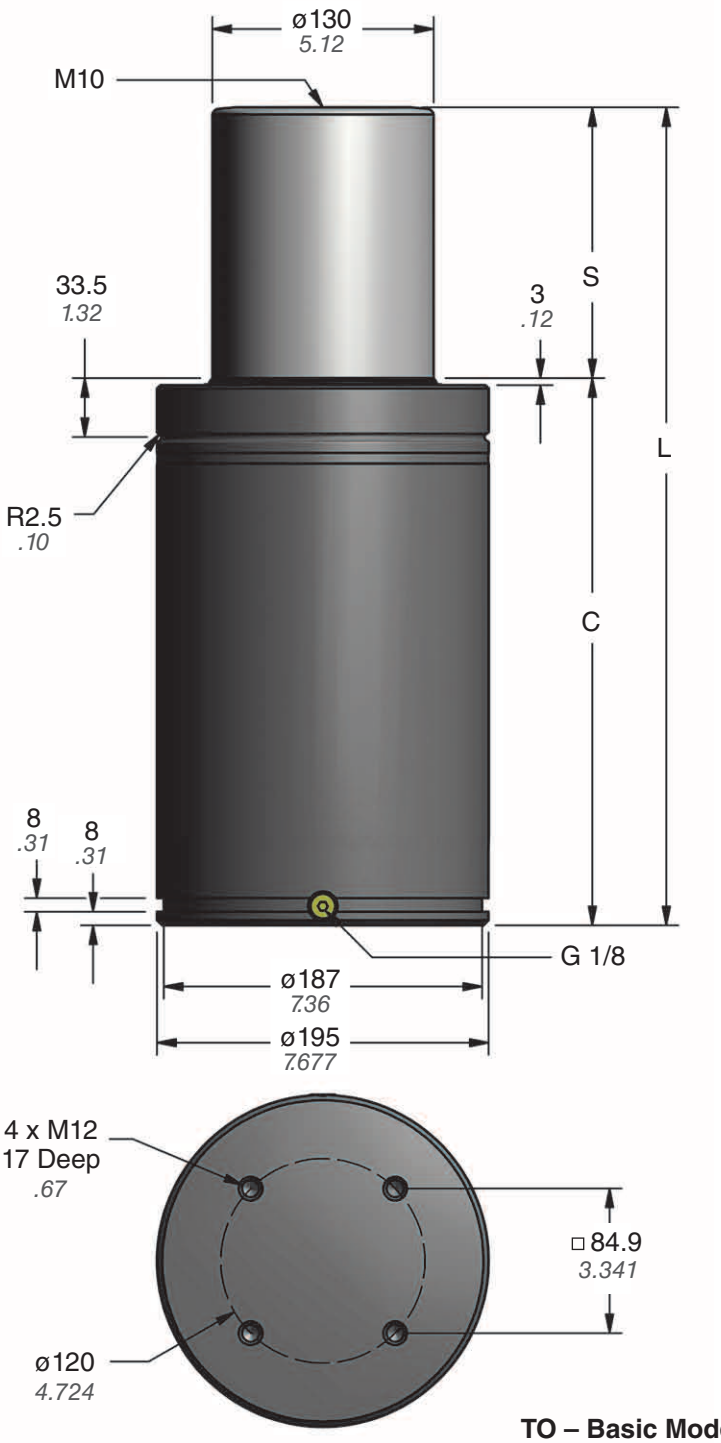
B319V 90.319V.9600

B19 Mount, see page 22.



DADCO offers a wide variety of mounts. If the mounts above are not suitable for the application, contact DADCO.

UX.20000 – 199 kN / 22 ton



TO – Basic Model

Part No.	S mm *inch*	C	L ±0.25 *±.010*
• UX.20000.025	25 *0.98*	185 *7.28*	210 *8.268*
UX.20000.038	37.5 *1.48*	197.5 *7.78*	235 *9.252*
• UX.20000.050	50 *1.97*	210 *8.27*	260 *10.236*
UX.20000.063	62.5 *2.46*	222.5 *8.76*	285 *11.220*
UX.20000.075	75 *2.95*	235 *9.25*	310 *12.205*
• UX.20000.080	80 *3.15*	240 *9.45*	320 *12.598*
• UX.20000.100	100 *3.94*	260 *10.24*	360 *14.173*
• UX.20000.125	125 *4.92*	285 *11.22*	410 *16.142*
UX.20000.150	150 *5.91*	310 *12.20*	460 *18.110*
• UX.20000.160	160 *6.30*	320 *12.60*	480 *18.898*
UX.20000.175	175 *6.89*	335 *13.19*	510 *20.079*
UX.20000.200	200 *7.87*	360 *14.17*	560 *22.047*
UX.20000.225	225 *8.86*	385 *15.16*	610 *24.016*
UX.20000.250	250 *9.84*	410 *16.14*	660 *25.984*
UX.20000.275	275 *10.83*	435 *17.13*	710 *27.953*
UX.20000.300	300 *11.81*	460 *18.11*	760 *29.921*

• Preferred Sizes

Ordering Example:

UX.20000.080. *TO.* *C.* *150*

Part Number:
Includes Series, Model and Stroke Length.

Mount Option:
TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

Mount Only Ordering Example: 90.21.10000

Charging Pressure:
15–150 bar *(220–2175 psi). When not specified, default is 150 bar.*

Operating System:
C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

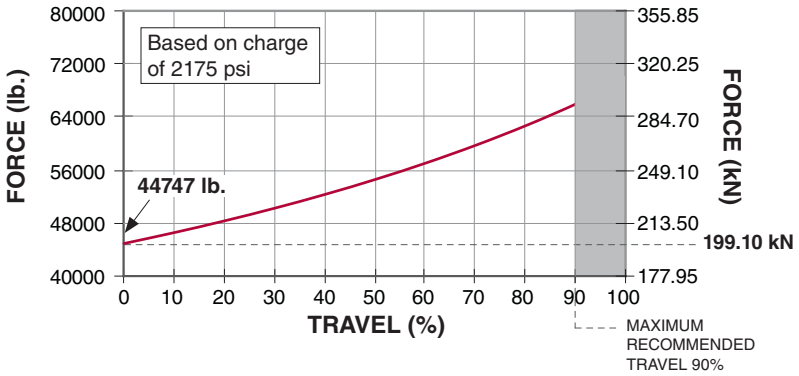
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UX.20000 – 199 kN / 22 ton

Force Charts

On-Contact Force

psi	lb.
2175	44747
2000	41147
1750	36004
1500	30860
1000	20574
500	10287
250	5143

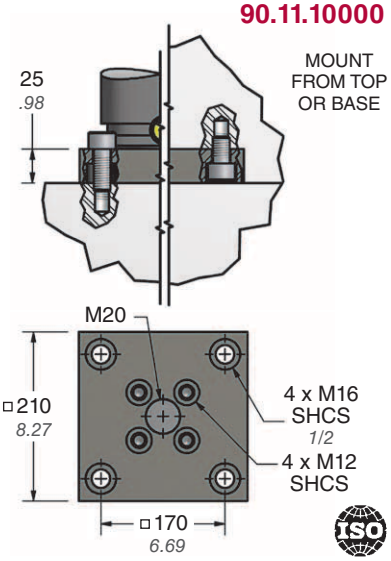


On-Contact Force

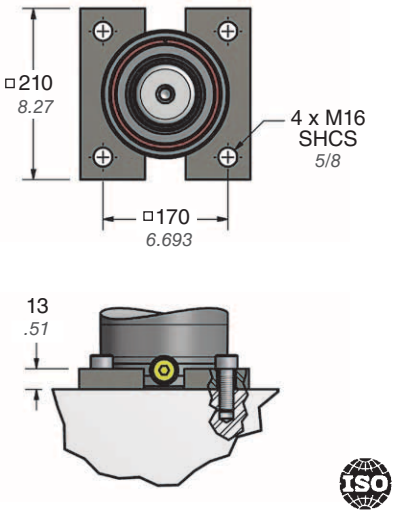
bar	kN
150	199.10
125	165.92
100	132.73
75	99.55
50	66.37
25	33.18
20	26.55

Mount Options

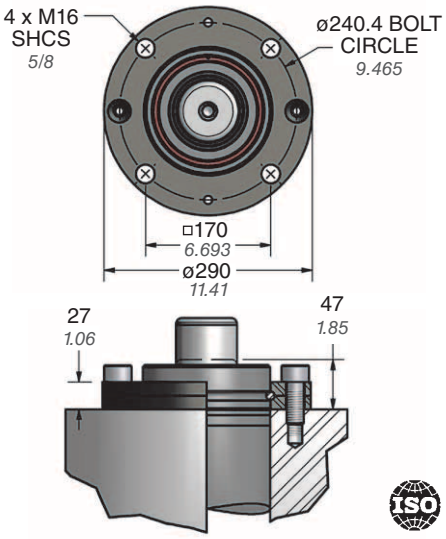
B11 90.11.10000



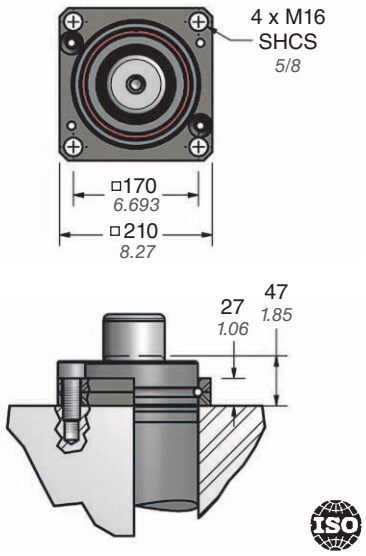
B12 90.12.10000



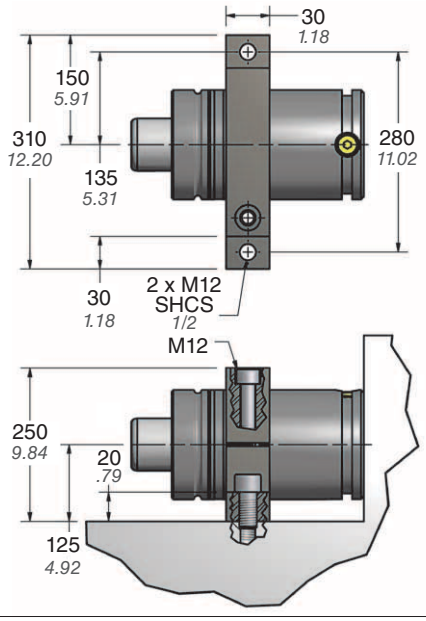
B21 90.21.10000



B25 90.25.10000

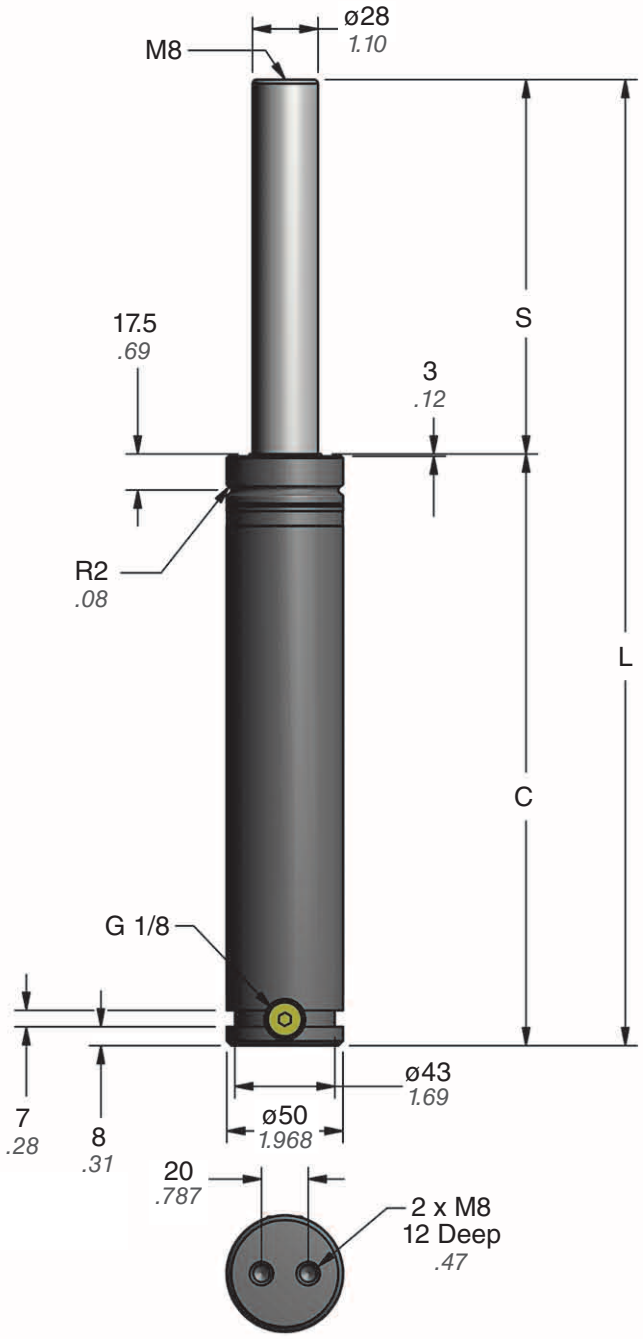


B19 90.19.10000



DADCO offers a wide variety of mounts. If the mounts above are not suitable for the application, contact DADCO.

UX.1000V – 9.2 kN / 1.0 ton



TO – Basic Model

Matches European Standards

Part No.	S mm *inch*	C	L ±0.25 *±.010*
UX.1000V.013	12.5 *.49*	107.5 *4.23*	120 *4.724*
UX.1000V.025	25 *0.98*	120 *4.72*	145 *5.709*
UX.1000V.038	37.5 *1.48*	132.5 *5.22*	170 *6.693*
UX.1000V.050	50 *1.97*	145 *5.71*	195 *7.677*
UX.1000V.063	62.5 *2.46*	157.5 *6.20*	220 *8.661*
UX.1000V.080	80 *3.15*	175 *6.89*	255 *10.039*
UX.1000V.100	100 *3.94*	195 *7.68*	295 *11.614*
UX.1000V.125	125 *4.92*	220 *8.66*	345 *13.583*
UX.1000V.160	160 *6.30*	255 *10.04*	415 *16.339*
UX.1000V.200	200 *7.87*	295 *11.61*	495 *19.488*
UX.1000V.250	250 *9.84*	345 *13.58*	595 *23.425*
UX.1000V.300	300 *11.81*	395 *15.55*	695 *27.362*

On-Contact Force

psi	lb.
2175	2076
2000	1909
1750	1670
1500	1432
1000	954
500	477
250	239

bar	kN
150	9.24
125	7.70
100	6.16
75	4.62
50	3.08
25	1.54
20	1.23

Ordering Example:

UX.1000V.080. TO. C. 150

Part Number:
Includes Series, Model and Stroke Length.

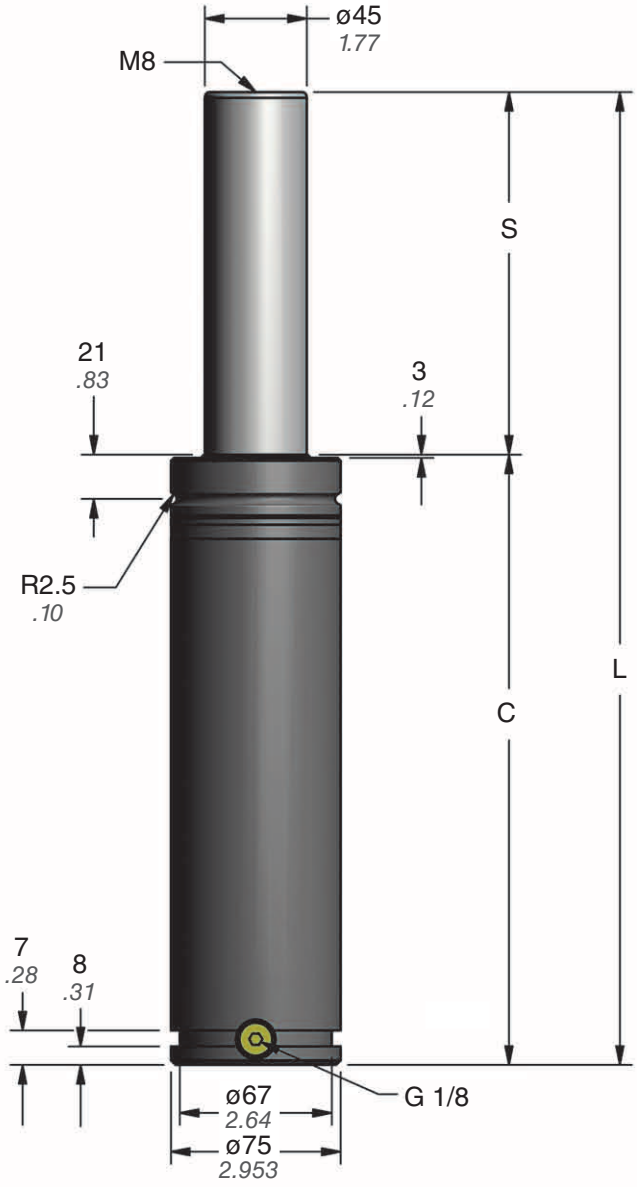
Mount Option:
TO = Basic Model.
Refer to page 7 for mount options. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

Charging Pressure:
15–150 bar *(220–2175 psi). When not specified, default is 150 bar.*

Operating System:
C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

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UX.2600V – 24 kN / 2.6 ton



TO – Basic Model

Matches European Standards

Part No.	S mm *inch*	C	L ±0.25 *±.010*
UX.2600V.025	25 *0.98*	135 *5.31*	160 *6.299*
UX.2600V.038	37.5 *1.48*	147.5 *5.81*	185 *7.283*
UX.2600V.050	50 *1.97*	160 *6.30*	210 *8.267*
UX.2600V.063	62.5 *2.46*	172.5 *6.80*	235 *9.251*
UX.2600V.080	80 *3.15*	190 *7.48*	270 *10.629*
UX.2600V.100	100 *3.94*	210 *8.27*	310 *12.204*
UX.2600V.125	125 *4.92*	235 *9.25*	360 *14.173*
UX.2600V.160	160 *6.30*	270 *10.63*	430 *16.929*
UX.2600V.200	200 *7.87*	310 *12.20*	510 *20.079*
UX.2600V.250	250 *9.84*	360 *14.17*	610 *24.016*
UX.2600V.300	300 *11.81*	410 *16.14*	710 *27.953*

On-Contact Force

psi	lb.
2175	5362
2000	4930
1750	4314
1500	3698
1000	2465
500	1233
250	616

bar	kN
150	23.86
125	19.88
100	15.90
75	11.93
50	7.95
25	3.98
20	3.18

Ordering Example:

UX.2600V.080. TO. C. 150

Part Number:
Includes Series, Model and Stroke Length.

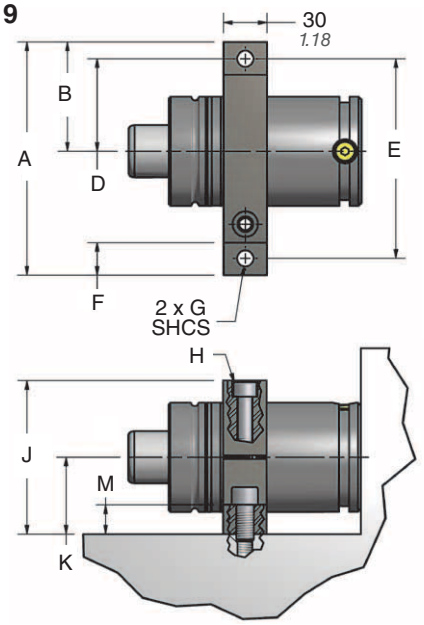
Mount Option:
TO = Basic Model.
Refer to page 11 for mount options. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

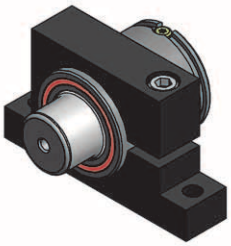
Charging Pressure:
15–150 bar *(220–2175 psi). When not specified, default is 150 bar.*

Operating System:
C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

Alternative Mount & Accessories

B19





SLIDE MOUNT INTO DESIRED LOCATION AND LOCK

90.19._____

Model	0800	1000	2600	4600	6600	9600
Part No.	00500	00750	01500	03000	05000	07500
A	100 *3.94*	130 *5.12*	160 *6.30*	195 *7.68*	220 *8.66*	260 *10.24*
B	46 *1.81*	60 *2.36*	75 *2.95*	92.5 *3.64*	105 *4.13*	125 *4.92*
D	37 *1.46*	50 *1.97*	63.5 *2.50*	80 *3.15*	92.5 *3.64*	110 *4.33*
E	82 *3.228*	110 *4.331*	137 *5.394*	170 *6.693*	195 *7.677*	230 *9.055*
F	18 *.71*	20 *.79*	22.5 *.89*	25 *.98*	27.5 *1.08*	30 *1.18*
G	M8 *5/16*	M8 *5/16*	M10 *3/8*	M12 *1/2*	M12 *1/2*	M12 *1/2*
H	M8	M8	M10	M12	M12	M12
J	60 *2.36*	80 *3.15*	105 *4.13*	125 *4.92*	148 *5.83*	200 *7.87*
K	30 *1.18*	40 *1.57*	52.5 *2.06*	62.5 *2.46*	74 *2.91*	100 *3.94*
M	15 *.59*	20 *.79*	20 *.79*	20 *.79*	20 *.79*	20 *.79*

Accessories

T-Handle
90.320.2 (M8: UX.0800 – UX.9600)
90.320.10 (M10: UX.20000)

To remove the piston rod when disassembling and position correctly when reassembling.



Port Servicing Tool
90.320.8

To perform all necessary servicing to the valve compartment.



C-Ring Removal Tool
90.355 (UX.0800-UX.4600)
90.356 (UX.1000-UX.20000)

To remove the C-Style Retaining Ring safely in a single controlled motion. For more information contact DADCO.



90.355



90.356

Valve Bleed Tool
90.360.4

Use the DADCO Valve Bleed Tool to slowly discharge a spring to the desired pressure.
For more information contact DADCO.



C-Ring Installation Tool
90.351.00500 (UX.0800)
90.351.00750 (UX.1000)
90.351.01600 (UX.1600)

To insert the C-Style Retaining Ring into the retaining ring groove.



90.351.01600

C-Ring Installation Tool
90.352 (UX.2600 – UX.9600)
90.352.10000 (UX.20000)

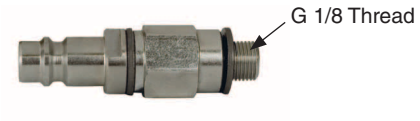
To insert the C-Style Retaining Ring into the retaining ring groove.



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Quick Disconnect Charging Nipple
90.310.111

Use the DADCO Quick Disconnect Charging Nipple to charge the UX Series gas springs. For more information, contact DADCO.



Removal Sleeve
90.340.00750 (UX.0800)
90.340.01200 (UX.1000)
90.340.01600 (UX.1600)
90.340.01500 (UX.2600)
90.340.03000 (UX.4600)
90.340.05000 (UX.6600)
90.340.07500 (UX.9600)



To position the cartridge below the C-ring groove when assembling or disassembling a gas spring.

DADCO Pressure Analyzer
90.315.5

DADCO's Pressure Analyzer includes two interchangeable bits to easily charge, discharge and gauge the pressure in any DADCO Nitrogen Gas Spring; request bulletin B01133D.



Portable Test Stand
90.305.3

Use the Portable Test Stand in conjunction with a Standard Load Cell for precise measurement of gas spring force. For use with UX.0800 – UX.6600 models only. For more information request bulletin 97B121.



Standard Load Cell

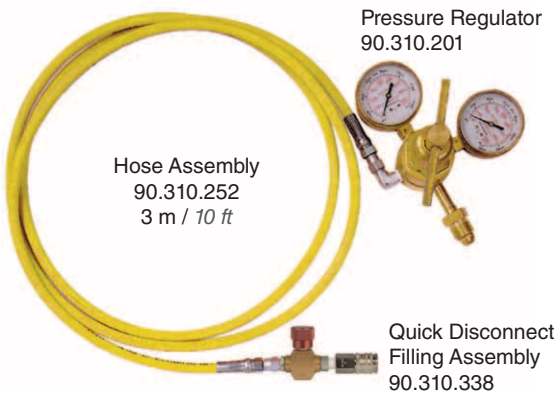
90.300.0750 (UX.0800)	**90.300.2600 (UX.2600)**	
90.300.1000 (UX.1000)	**90.300.6600 (UX.6600)**	**90.300.9600 (UX.9600)**
90.300.01500 (UX.1600)	**90.300.4600 (UX.4600)**	**90.300.20000 (UX.20000)**

The Standard Load Cell gives precise measurement of gas spring charging pressure. Each model requires its specified load cell; request bulletin 97B119G.



Quick Disconnect Charging Assembly
90.310.040

Use the DADCO Quick Disconnect Charging Assembly, *90.310.040*, with the *90.310.111* Charging Nipple or *90.315.5* Pressure Analyzer to charge self-contained gas springs. The *90.310.040* can also be used with a DADCO control panel to charge linked systems. The *90.310.040* includes the *90.310.201* Pressure Regulator, *90.310.252* Hose Assembly and the *90.310.338* Quick Disconnect Filling Assembly.



UX Series Repair Kits

The UX Series repair kit includes a fully assembled cartridge, dust cover, bottle of assembly oil and maintenance manual.

Model	Kit Number	Model	Kit Number
UX.0800	90.108.00800	UX.2600V	90.108V.02600
UX.1000	90.108.01000	UX.4600	90.108.04600
UX.1000V	90.108V.01000	UX.6600	90.108.06600
UX.1600	90.108.01600	UX.9600	90.108.09600
UX.2600	90.108.02600	UX.20000	90.108.20000

Ultra Force Extended® Technical Data

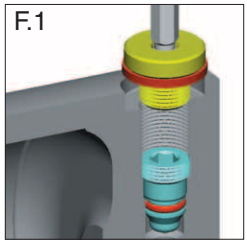
Operating Specifications

Charging Medium:	Nitrogen Gas
Charging Pressure Range:	15 – 150 bar (*220 – 2175 psi*)
Operating Temperature:	-6°C – 71°C (*20°F – 160°F*)*
Maximum Speed:	.5 m/sec (*20 in/sec*)

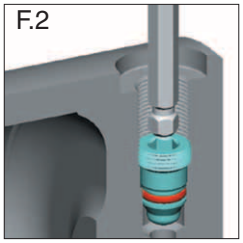
**CAUTION!* For extended range applications contact DADCO.*

CAUTION! DO NOT attempt maintenance on spring until internal pressure is exhausted.

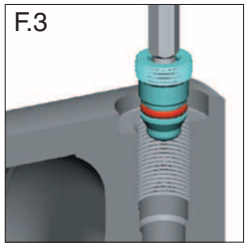
F.1



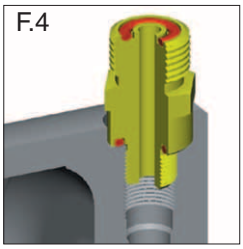
F.2



F.3



F.4



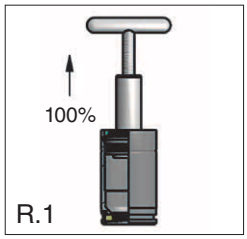
Self-contained to Linked Operation

- Always wear safety glasses when maintaining gas springs. When exhausting pressure, place the gas spring horizontally with the port up for safety.
- Remove port plug, *90.505.110* (F.1).
- With the cylinder in the horizontal position, depress the valve stem using the appropriate tool (F.2). Cover the port with a cloth to absorb discharge.
- After all of the gas pressure is exhausted, be sure that the piston rod will retract into the tube manually. If not, try depressing the valve again. If still unsuccessful, stop and contact your DADCO Service Representative.
- Remove the valve using the appropriate tool (F.3). Lubricate the threads and o-ring of the port adapter being installed (F.4).
- Install port adapter in open port (F.4). A wide variety of port adapters are available, refer to DADCO's Nitrogen Gas Spring Linked System Catalog.

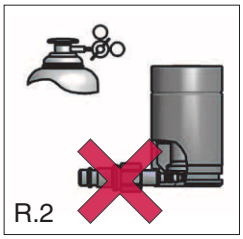
Recharging Self-contained Gas Spring

- Hold the spring vertically at all times during filling. Never compress the gas spring in a vice or clamp outside of the die or application as damage to the gas spring can result.
- Never fill a gas spring unless the rod is in the fully extended position (R.1). Filling a gas spring with its rod down can result in improper retaining ring seating (R.2). Thread the T-handle, *90.320.2 or 90.320.10,* into the rod end and depress the valve stem with the appropriate tool, then pull the rod cartridge assembly up until it is seated firmly against the retaining ring (R.1).
- Charge the gas spring to the desired pressure. Refer to DADCO's UX Series Maintenance Instructions for complete step-by-step instruction.

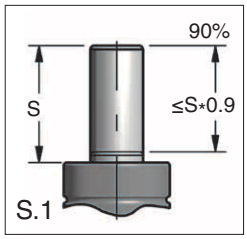
R.1



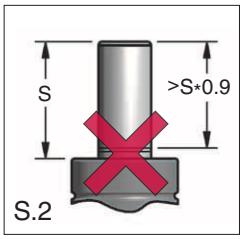
R.2



S.1



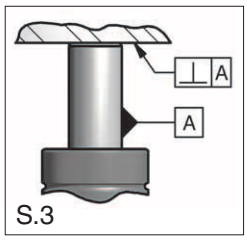
S.2



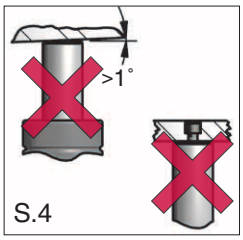
Provide Stroke Reserve

- DADCO's UX Series Gas Springs will permit travel of the full nominal stroke; however, at least a 10% stroke reserve is recommended to achieve optimal performance and safety (S.1, S.2).
- Overstroking the rod or impacting the top of the gas spring will cause permanent damage.

S.3



S.4



Avoid Side Loading

- Side loading resulting from press action or die construction causes increased wear on bearing, seal and piston rod (S.4). Therefore, avoid side loading when possible (S.3).

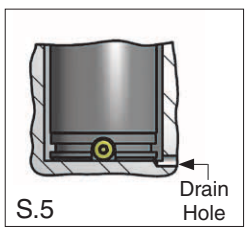
Rod End Thread

- The end of the piston rod has a construction thread intended for assembly and disassembly only and should never be used to mount or secure the gas spring (S.4). Die vibration and/or misalignment will damage the spring.

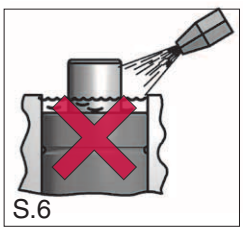
Protect From Fluids

- Direct contact with certain die lubricants and cleaners should be avoided (S.6). Protect gas springs by providing adequate drainage in gas spring pockets (S.5).

S.5



S.6



DADCO®
The global leader in nitrogen gas spring technology
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